



Supplementary Materials for

Ten millennia of hepatitis B virus evolution

Arthur Kocher *et al.*

Corresponding authors: Arthur Kocher, kocher@shh.mpg.de; Johannes Krause, krause@shh.mpg.de;
Denise Kühnert, kuehnert@shh.mpg.de

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Materials and Methods

HBV Detection in ancient individuals

Shotgun sequencing data obtained from ancient human remains in the context of projects conducted at the Max Planck Institute for the Science of Human History were screened for traces of HBV DNA using MALT ver. 0.3.8 (59), with a reference dataset representative of known modern and ancient HBV diversity as well as other orthohepadnaviruses (Supplementary Methods). DNA vector sequences containing Woodchuck Hepatitis Virus or HBV Posttranscriptional Regulatory Elements were also included in the reference dataset to avoid false positive detection due to these potential laboratory contaminants. Reads mapping specifically to HBV were extracted from the MALT results (rma6 files) for further examination using Maltextract (part of the HOPS pipeline (60)) with a minimum percentage identity ($-\text{minPI}$) of 80, a top percent value ($-\text{top}$) of 0.05 and all other parameters set to default.

DNA capture and sequencing

Approximately 4% (325/7918) of screened datasets showed putative traces of HBV DNA. HBV-DNA enrichment using in-solution capture was performed for candidate DNA libraries prepared from 136 Eurasian and American individuals dated between ~10.5 and 0.4 ka. (Data S1). Among individuals for which the age and sex information was available, 67% were adults (80/119), and 56% (67/120) were males. For some of the individuals, several libraries were included, resulting in a total of 154 libraries (Data S1). 104 of these were prepared from teeth, while 50 were prepared from bones, including 32 from the petrous bone. HBV probe sequences were designed based on the reference dataset used for screening with a 1-bp tiling, a length of 52 bp and an additional 8-bp linker sequence (CACTGCGG), as previously described (12, 61). Low-complexity and duplicated probes were removed, resulting in 221,190 unique probe sequences. The set was quadrupled to fill the feature space of an Agilent SureSelect DNA Capture Array and turned into an in-solution DNA capture library (61). Candidate libraries were reamplified with IS5/IS6 primers and Herculase II Fusion DNA Polymerase to reach a concentration of ~400 ng/ μL .

as measured on a NanoDrop 8000 spectrophotometer (Thermo Fisher Scientific Inc.; Waltham, MA). Capture was performed on 5.25 μ L of each library. After two consecutive rounds of enrichment, a reconditioning PCR was made to eliminate heteroduplexes originating from mixed-template PCR products. Enriched libraries were then pooled equimolarly (10 mM final concentration), and prepared for shotgun sequencing on an Illumina HiSeq 4000 Systems platform (Illumina, Inc., San Diego, CA) with 2x75 paired-end cycles.

Sequencing data preprocessing

Raw sequencing reads were assigned to corresponding libraries based on sequenced P7 and P5 indices (allowing for one mismatch per index). Adapter trimming and paired-read merging were then performed using *AdapterRemoval* 2.3.0 (62). Fragments smaller than 30 bp or containing more than three unresolved nucleotides ('N') were excluded. In order to limit the impact of inter-sample contamination due to index-jumping (63), we filtered sequencing reads based on their copy number similarly to Esling *et al.* (64). In brief, the rationale was that reads originating from authentic DNA molecules amplified in a given library should exhibit higher duplication levels than spurious reads arising from molecular and sequencing artefacts. For each library, we recovered reads carrying one of the two library indices, but for which the second index resulted in a combination that wasn't used in any of the sequenced libraries (i.e. necessarily resulting from index jumping). Reads assigned to the given library were kept only if their copy number was significantly higher than those recovered with unused index combinations, as assessed by the modified Thompson τ -test for outlier detection (64). After DNA damage assessment, one bp was clipped from both ends of reads. Resulting files were then combined with HBV reads recovered from shotgun sequencing of the same libraries, and merged for each individual.

HBV genome assembly

Due to evolutionary divergence, ancient HBV genomes may significantly differ from modern ones. Because this could hamper the reconstruction of ancient HBV genomes through a reference-mapping strategy, we first performed *de novo* assembly of the reads with Novoplasty (65), using a genome range of 3000 to 3500 bp, a k-mer length of 24 and

the single-end read option (reads were already merged at this step of the analysis). When a complete circularized assembly could not be obtained, we mapped reads against the HBV reference genome (GenBank accession: NC_003977) within *Geneious* 9.1.8., allowing for 15% mismatches and 10% gaps. We used a fine tuning with 5 mapping iterations against the temporary consensus, which allows recovering highly divergent regions. Alignments were then visually inspected and amended if necessary. In particular, we checked for the consistency of reading frames within each gene and for alignment issues around large indels. Preliminary consensus sequences were produced from these alignments using a 1x coverage threshold and a 50% majority rule.

Resulting *de novo* assemblies and preliminary consensus sequences were then used as references for corresponding sequencing datasets within the *EAGER* pipeline (66), in order to produce final conservative consensus sequences as well as coverage and damage statistics: reads were mapped using the CircularMapper option with a mismatch parameter (-n) of 0.01, a quality filtering (-q) of 30 and an elongation factor of 500. Mapped reads were deduplicated with *Markduplicates* (part of the *Picard* tools; <http://broadinstitute.github.io/picard/>) and realigned around indels using the *GATK toolkit* (67). Resulting binary alignment map (BAM) files were then used to produce tables detailing the number and nucleotide composition of reads covering each genomic position with *pysamstats* (<https://github.com/alimanfoo/pysamstats>). Finally, we called consensus sequences for each genome using a 3x coverage threshold with a 90% majority rule. The same procedure was used to assemble an HBV genome recovered from the Loschbour individual's shotgun sequencing data (38), and to reanalyse the data previously published by Krause-Kyora *et al.* (13).

In order to assess the effect of the skeletal element (bone vs. tooth) and library protocol (single-stranded vs. double-stranded) on the success of HBV genome reconstruction, we used a linear mixed-effect model as implemented in the *R* package *lme4* (68). Mean HBV coverage was used as the response variable after determining the optimal Box-Cox transformation. Because in some cases, several libraries were sequenced from the same individual, a full interaction mixed effect model with the individual as a random effect was then fitted. Starting with the full-interaction model, backward model-selection was

performed using ANOVA to find the simplest model which still adequately described the data. Marginal means were estimated to assess the relative effect of each level within predictive variables, and the significance of observed differences, while still accounting for random effects. The analysis was repeated using a skeletal element variable in which the petrous bone was distinguished from other bones. Finally, we used the same approach to assess whether a single-stranded library protocol would reduce the difference of coverage obtained between the double and single-stranded regions of the HBV genome (fig. S9B). In that case, the ratio of mean coverage obtained in these two regions for each genome was used as the response variable. Although its precise location along the genome may vary, the single-stranded region was considered to be located between position 750 and 1600 for this analysis (69). Genomes for which the single-stranded region had a coverage of 0 were excluded from the analysis.

DNA damage assessment

Deamination profiles typical of ancient DNA damage were investigated. For some individuals, we captured several DNA libraries that had not necessarily undergone the same uracil-DNA-glycosylase (UDG) treatment (70). Damage assessment was therefore done separately for each library. Untrimmed reads were input in *EAGER* as described above, and the pipeline was complemented with DNA damage assessment using *mapDamage* (71). The level of damage on human DNA was assessed in each library using the same pipeline with the GRCh37/hg19 Human reference genome. The correlation between damage on HBV and human DNA was assessed using a linear regression of the C to T substitution frequency at the 5' end of human and HBV-mapping reads respectively (excluding UDG-full libraries and libraries from which less than 100 reads mapping to HBV were found after filtering and deduplication; fig. S9F).

Identification of mixed HBV infections

To assess the presence of mixed HBV infections in some individuals, we looked for heterozygous-like signals along the HBV genome sequence in sequencing read alignments (Data S2). Our rationale was that in case of a mixed infection, one should observe more frequent heterozygous-like positions than expected with molecular and sequencing

artifacts only. For each sample, we computed the relative support of the two major variants at each HBV genomic position, for which we also derived a 95% confidence interval using Wilson's method (as implemented in the *R* package *binom*). We then defined heterozygous-like positions as positions being covered at least 10x and for which the support of both major and second major variants was significantly below 90% and above 10% respectively. Cases where the major variant was C or G and the second variant was T or A, respectively, were excluded, in order to discard any heterozygous-like signal possibly due to DNA damage. The baseline probability of a position to be detected as heterozygous-like was estimated as the overall fraction of heterozygous-like positions over positions covered >10x in the complete dataset. A given sample was considered as an outlier and indicative of a mixed infection if it contained a number of heterozygous-like positions that exceeded the 95% quantile of a binomial distribution parameterized with the above-mentioned baseline probability and the number of positions covered >10x in the given sample as number of trials. The results were identical when using the 99% quantile instead. Additionally, we visually inspected read alignments around the 5' end of the C gene. Cases where only a subset of the reads carried a ~40-bp-long insertion (similar to the one found in genotype G) were also considered as evidence of mixed infections. The consensus sequence obtained from each sample showing signs of mixed HBV infections was expected to represent the genome sequence of the most abundant strain. We also tried to estimate to which lineage the minor strain belonged using the *EPA-ng* algorithm [(72); see below]

Genomic dataset and sequence alignment

For phylogenetic analyses, we gathered a set of modern HBV genomes encompassing the currently described diversity of the virus (Supplementary Methods) to which we added previously published ancient HBV genomes (11–16) as well as the genomes generated in this study for which at least 50% of the sequence was resolved (105/137). Sequences were aligned using *MAFFT* v7.475 (73) with the iterative refinement method for global alignments. The resulting alignment was inspected using *Geneious* and corrected around large indels when necessary. Highly divergent and potentially misaligned regions, as well as regions containing indels, were masked from the alignment using *Gblocks* (74) with default parameters. An additional stretch of 9 nucleotides flanking the location of large

insertions in the preS1 region (pos. 2990-2998) appeared potentially misaligned and was masked as well.

Temporal signal assessment and phylogenetic analyses

The HBV genome alignment was used to construct a phylogenetic tree with *RAxML* v. 8.2.12 (75), using the GTRCAT substitution model and the rapid bootstrap algorithm with the autoMRE bootstopping criterion. The resulting ML tree was midpoint-rooted, and the temporal signal of the genomic dataset was assessed using a linear regression of root-to-tip genetic distances (measured in substitutions/site) against genome sampling dates (76). Root-to-tip regression indicated that our dataset exhibited a strong temporal structure (fig S3), which was further confirmed by a formal Bayesian model comparison [(77); see below].

We therefore performed a time-calibrated phylogenetic analysis using *BEAST2* (78). Genome ages were used as calibration points (using the midrange of C14 or archeological dating, and 0 for modern genomes). In order to choose the most appropriate tree prior and clock model, we performed model selection using path sampling (79) as implemented in the *model-selection* package. We compared constant coalescent, exponential coalescent, Bayesian skyline coalescent (80) (with either 5 or 10 population size groups), and birth-death skyline (81) tree priors, each of which were combined with either a strict or a relaxed lognormal clock model (82). For coalescent tree priors, the upper bound of the population size was set to 38M (default value increased by two orders of magnitude). For the birth-death skyline model, we allowed 5 shifts of the reproductive number, for which we used a lognormal prior with mean=0 and sd=0.5. We considered a constant rate to become non-infectious with an exponential prior with mean=0.01, and a constant sampling proportion with a uniform prior between 0 and 1. Additionally, we parameterized the model using a sampling probability ρ at time 0 with a uniform prior between 0 and 1. For each of these models, we used a GTR+GAMMA+I substitution model with four gamma categories. A uniform distribution between 10^{-9} and 10^{-3} subst.site⁻¹.year⁻¹ was used as a prior for the mean clock rate, based on the range of previous estimates (10, 14). The weights of tree-related operators were increased three-fold in order to improve mixing of the tree topology.

All other settings were left at their default values (*BEAUTi* v. 2.5.2). Path sampling was run for each of these models using 16 steps of 80M MCMC iterations and 50% burn-in. Resulting marginal likelihood estimates were used to compare the fit of each model. We then used path sampling in the same way to perform Bayesian evaluation of the temporal signal (77). The previously selected model was modified to drop the time-calibration information: tip dates were not used and the clock rate was fixed to 1. The use of authentic genome dates yielded a better model fit, indicating significant temporal signal (log Bayes Factor: 338).

The selected model was used for phylogenetic inference in *BEAST2*. MCMC sampling was performed using 500M iterations including 10% burn-in, or more when necessary. We assessed convergence by ensuring that effective sample size was above 200 for each parameter. A maximum clade credibility (MCC) tree was derived from the posterior tree distribution using *TreeAnnotator* (83). We then explored plausible alternative topologies for two deep nodes which had relatively low support in the resulting MCC tree: the Eurasian branch and the clade corresponding to the Eurasian branch with the exclusion of the Mesolithic 1 and Southeast Asian NHP clades. We retrieved MCC trees conditioned on the absence of these respective clades (i.e. based on a tree sample in which all trees containing these clades were filtered out). We then used the full posterior tree sample to compute node posterior supports on these alternative topologies.

In order to evaluate the robustness of our results with respect to different aspects of the dataset and model assumptions, we performed a series of sensitivity analyses. First, because genetic recombination might interfere with phylogenetic inference, we performed an analysis using an alignment in which recombining regions identified with *RDP4* (84) were masked from recombinant genomes. Second, because mixed HBV infections might result in genome assembly issues such as the reconstruction of artificial hybrids, we performed an analysis excluding all ancient genomes recovered from individuals in which mixed HBV infections were detected. Lastly, because it is frequently observed that evolutionary rates may vary depending on the considered time-scale (85), we performed an analysis using a time-dependent-rate model as implemented in *BEAST* v.1.10.4 (86, 87). We used a five-epoch model with transitions every 2,000 years, and the midpoint of

each epoch as a reference point to compute the epoch's rate (taking 9 ka for the last epoch). We used Normal priors with mean=-11.5 and 0 and sd=5 and 5 for the intercept and slope coefficients of the log-linear relationship between the substitution rate and time, respectively (corresponding to an expected rate of $\sim 10^{-5}$ subst.site⁻¹.year⁻¹ at time 0 and no prior assumption on the direction of the relationship). Other model specifications were identical to the main analysis. Additionally, we assess the performance of this model compared to previously tested clock models using path sampling as previously described using BEAST v.1 (88).

Genomic characteristic and genotyping of ancient HBV strains

Examination of inter-strain genetic distances highlighted the difficulty to consistently classify ancient strains into genotypes using the convention used for modern ones (*i.e.* an average genetic distance threshold of 7.5%). Indeed, some ancient genomes showed less than 7.5% divergence with modern strains from several genotypes, and the use of this threshold would categorize ancient strains into groups that were not necessarily recovered as monophyletic (table S4; supplementary text). Therefore, ancient strains grouped monophyletically and showing less than 7.5% genomic divergence with modern strains from a given genotype were classified as such. Otherwise, we used a classification scheme based on well-supported phylogenetic clades that seem to have a consistent spatiotemporal distribution: Mesolithic lineages 1 and 2, Anatolian Early Neolithic lineage, Western-Eurasian Neolithic-to-Bronze-Age lineage, Ancient American lineages 1, 2 and 3. Finally, we inferred the serotype of ancient strains based on their surface protein sequence (89), assessed their HBeAg status based on their pre-core region sequence, and looked for previously described or unknown genomic indels (3) by visually inspecting the sequence alignment (fig. S8, Data S1).

Phylogenetic analysis

We explored the transmission dynamics of WENBA HBV strains in Eurasia using a birth-death skyline model as implemented in BEAST2 (81) with a dataset consisting of only WENBA strains. When several genomes were recovered from the same site and appeared closely related based on the initial phylogenetic analysis, only one genome was kept (the

most resolved one), in order to reduce sampling bias (see Supplementary Text for the list of excluded genomes). The dataset was complemented with 35 sequences of modern genotype G genomes taken from a previous phylodynamic analysis [(90); see Supplementary Text for a list of accessions]. Sequences were aligned and highly divergent regions were masked as previously described. Because of the restricted dataset employed for this analysis, a strict clock model was used, with a uniform prior between 10^{-9} and 10^{-3} .subst.site⁻¹.years⁻¹ for the substitution rate. Based on the assumption that the majority of HBV infections are long-lasting chronic infections, we used a constant rate to become non-infectious with a lognormal prior with mean=0.05 (in real space) and sd=0.1, corresponding to a 95% percentile interval for the average duration of infectivity between ~15 and ~25 years. For the diversification/extinction ratio (reproductive number) and the sampling proportion, we allowed three time shifts (i.e. four parameter values), and (i) a lognormal prior with mean=3 (in real space) and sd=1 for the diversification/extinction ratio and (ii) a uniform distribution between 0 and 1 for the sampling proportion. Time shifts were estimated within three equally-spaced time windows between 0 and 9 ka (allowing a fourth shift did not appear to yield additional information). The posterior distribution of time shifts and piecewise-constant diversification/extinction ratios were used to derive the posterior distribution of the diversification/extinction ratio through time using a grid of 100 years.

Lineage assignment of low-coverage genomes

We used the *EPA-ng* algorithm (72) as described previously (91) to perform phylogenetic placement of low-coverage HBV genomes that were not included in phylogenetic analyses. Consensus sequences were obtained for low-coverage genomes using the previously described approach modified with a 1x coverage threshold. Resulting sequences were aligned with the high-coverage-genome alignment (which was used for phylogenetic analyses) using *MAFFT* with the *-add* and *-keeplength* options to preserve the original alignment structure. The resulting alignment was corrected around large indels when necessary, and potentially misaligned regions were masked as previously described. The high-coverage-genome alignment was used as a reference alignment, and the low-coverage-genome alignment was used as a query for *EPA-ng*. The reference tree and

substitution model specifications were taken from the previously conducted ML phylogenetic analysis. The taxonomic assignment method of *Gappa* (92) was used to assign genotypes or previously defined ancient HBV lineages to each low-coverage genome. We used a taxonomic reference file in which all reference sequences (highly-covered genomes) were associated with a taxonomic path consisting of “Hepatitis B virus” followed by the corresponding genotype or ancient lineage (i.e. we did not allow subgenotype nor intermediate level assignments). For each low-coverage genome, the most supported assignment was identified using likelihood weight ratios (LWRs).

Lineage assignment of minor strains within mixed infections

For samples from which mixed infections were identified, we tried to infer the lineage to which the minor strain belonged based on identified genomic polymorphisms. We used the python package *pysam* (<https://github.com/pysam-developers/pysam>) to create new BAM files containing only the sequencing reads carrying minor variants at identified heterozygous-like positions, with the additional condition that the variant support was significantly below 40%. This was to avoid selecting mixtures of reads corresponding to several strains (in case those were found in even proportions), and left 12 samples for which such reads could be retrieved and analyzed. Consensus sequences were obtained from these alignments using a 1x coverage threshold, and used to perform phylogenetic placement with *EPA-ng* as previously described (table S3).

Detection of recombinant genomes

RD4 (84) was used to detect signals of genetic recombination in our dataset. A complete exploratory search was run on the sequence alignment used for phylogenetic analyses with the seven integrated methods employed by default. Events recovered by all methods were further examined iteratively, starting with the most supported ones. Based on the RDP plot and the topology of local ML trees (constructed with the minor and major parental regions of the alignment), the event was either confirmed or rejected. More specifically, we expected that a group of genomes identified as resulting from the same recombination event should form a monophyletic clade closely related to respective parental genomes in local ML trees. In some cases, swapping parental or recombinant genomes in the identified

triplet resulted in more likely scenarios. Examination of local ML trees sometimes revealed that close relatives of identified recombinant(s) were affected by the same event, in which case they were marked as such. When a recombination event was confirmed, we accepted it and performed a rescan of the dataset. The procedure was repeated until no further unaccepted event was recovered by all methods.

Human population genetics

Human genome-wide data for some of the ancient individuals from which HBV genomes were recovered was taken from the latest Allen Ancient DNA Resource (<https://reich.hms.harvard.edu/allen-ancient-dna-resource-aadr-downloadable-genotypes-present-day-and-ancient-dna-data>) V44.3 January 20 2021. Principal components analysis (PCA) was conducted using *smartpca* v16000 from the *EIGENSOFT* package (<https://github.com/DReichLab/EIG>). The principal components were calculated on a set of modern west Eurasian individuals genotyped on ~597000 sites from the Affymetrix Human Origins array (38, 93–96). Ancient individuals were projected onto the resulting principal components (lsqproject: YES) and shrinkmode was used to reduce the effect of modern populations “stretching” of axes (shrinkmode: YES).

Supplementary Text

List of GenBank accessions of HBV genomes used for the screening of shotgun sequencing data and probe design

We included a wide range of HBV genomes encompassing all known modern HBV genotypes as well as ancient genomes and orthohepadnaviruses found in non-human mammals: LT992459, LT992455, LT992454, LT992448, LT992447, LT992444, LT992443, LT992442, LT992441, LT992440, LT992439, LT992438, JN315779, MG585269, AB076679, AB116084, AB453988, AY738142, GQ477499, AY934764, FJ692556, FJ692598, FJ692611, GQ161813, GQ331046, AB073858, AB033555, AB219429, AB219430, AP011089, AB073835, AB287316, AB287318, AB287320, AB287321, DQ463789, DQ463792, AB241117, DQ993686, AB111946, AB112066, AB112472, DQ089767, X75656, X75665, AB048704, AB048705, AF241411, AP011100, AP011102, AP011103, AP011106, AP011108, FJ899792, JN642140, GQ477453, GQ477455, JN642160, JN642163, JN688710, JN688711, GQ922005, HE974378, KJ470893, KJ470896, KJ470898, FJ904430, FJ904436, AB033559, AB048701, AB048702, AB188243, AB210818, AM494716, AY796031, AY902768, DQ315779, X80925, X75657, X75664, AY090458, AB116654, FJ657525, AY090455, AY311369, DQ899144, DQ899146, AB116549, X75663, AF223962, AB166850, AB056513, AB064312, AF405706, AB059660, AB375163, AY090454, AY090457, AB486012, AY330911, AJ131571, AY781180, U46935, AJ131567, AF193863, EU155824, KC790378, KC790377, KC790376, NC_001484, U29144, NC_004107, AF046996, KY703886, MH307930, AF498266.

We also included the three ancient genomes recovered by Krause-Kyora *et al.* (13), as well as DNA vector sequences containing Woodchuck Hepatitis or HBV posttranscriptional regulatory elements to avoid false-positive detection due to these potential lab contaminants (those were excluded for probe design): GQ202119, GQ202120, DQ869006, JN898962, JN898959, KJ796484, AB902850, KJ411917, KJ411912, KJ411915, KJ411918, KJ411916, KJ411919, KJ411911, KT345943, JX006096, JF927991, KX757240, KX757239, KX757255, KX757254, KX757253, KX757252, KX75725

1,KX757250,KX757249,KX757248,KX757244,KX757247,KX757246,KX757245,X757242,KX757243,KX757241,KX757238,FR822201,JX861384,KM519787,KM519788,JN622008,EF205034,EF205035,KJ697750,KJ697752,KJ697753,KJ697751,HG530137,GQ872121,KC152483,KC152484,KC152485,KC152482,KC152481,JQ086322,KF486506,FJ797421,EF177827,AY468486,KC262216,GU253315,GU253314,GU253312,GU253313,EU048697,EU048696,EF186079,EU000249,EF186078

List of GenBank accessions of modern HBV genomes used for phylogenetic analyses

Genotype A: AB076679, AB116084, AB453988, AY738142, GQ477499, AY934764, FJ692556, FJ692598, FJ692611, GQ161813, GQ331046

Genotype B: AB073858, AB033555, AB219429, AP011089, AB073835, AB287316, AB287318, AB287320, AB287321, DQ463789, DQ463792, AB241117, DQ993686, FJ899779

Genotype C: AB111946, AB112066, AB112472, DQ089767, X75656, X75665, AB048704, AB048705, AF241411, AP011100, AP011102, AP011103, AP011106, AP011108, AB049609

Genotype D: NC_003977, FJ899792, JN642140, GQ477453, GQ477455, JN642160, JN642163, JN688710, JN688711, GQ922005, HE974378, KJ470893, KJ470896, KJ470898, FJ904430, FJ904436, AB033559, AB048701, AB048702, AB188243, AB210818, AM494716, AY796031, AY902768, DQ315779, X80925, FJ904399, AY721612, AY741797, AB270543, EU594409, AB109476, AB555496, GQ205377

Genotype E: X75657, X75664, HM363593

Genotype F: AY090458, AB116654, FJ657525, AY090455, AY311369, DQ899144, DQ899146, AB116549, X75663, AF223962, AB166850, JN792922

Genotype G: AB056513, AB064312, AF405706

Genotype H: AB059660, AB375163, AY090454, AY090457

Genotype I: AB562463, FJ023669, EU835241

Genotype J: AB486012

Genotype cpz: AY330911, AJ131567, AF498266

Genotype gbn: AJ131571, AY781180, U46935

Genotype oru: AF193863, EU155824

List of genomes used for the phylodynamic analysis

WENBA HBV strains included in the main phylogenetic analysis were used for the phylodynamic analysis. When several genomes were recovered from the same site and appeared closely related based on the initial phylogenetic analysis, only one genome was kept (the most resolved one), in order to reduce sampling bias. Thus, the following genomes were excluded: I0411, I0100, I2005, I2020, I2031, MK5009, MKL025, MIS002, WEH008.

We further included the following modern genotype G genomes from (52): AB064310, AB064311, AB064312, AB064313, AB625342, AF405706, AP007264, DQ207798, EF634480, GU563556, GU563559, GU565217, HE981171, HE981172, HE981174, HE981175, HE981176, KF414679, KF767450, KF767451, KF767452, KR230749, KX264500, KY004111, KY004112, AB375170, AB375169, AB375167, AB375166, AB375168, AF160501, AB056513, AB056514, AB056515, JQ707668.

We excluded seven sequences from the original dataset that had length incompatible with functional open reading frames, pointing to assembly issues: EF634481, HE981173, KF779233, KF779235, KF779357, KF779267 and JF439787 (KF779233, KF779235 and KF779357 also contained a highly divergent sequence at the end of the X gene). We also excluded sequence AB625343 which contained a sequence at the beginning of the S gene that appeared similar to that found in genotype H.

Archaeological background

Here we provide information on the archaeological context for some of the samples analyzed in this study (for the rest of the samples, see references in Data S1).

Minino 2, Vologda region, Russia (MN2003): The archaeological complex of Minino comprises artifacts and human remains from two Mesolithic-Early Neolithic cemeteries (Minino I and II), separated by a distance of 230 m, near the lake Kubenskoe in the Vologda region, Russia. **MN2003** (Grave VI/ind. 1) was found in a grave from Minino II, which was initially identified as a single grave by the archaeologists, although it contained remains from two individuals: an adult male (**MN2003**), for which an almost complete skeleton was found, and another adult individual, from which only a few skeletal elements were recovered. This combination of complete and partial skeleton remains within the same grave had never been observed before in the site, but later findings revealed that it was a common feature of Minino (half of the graves in Minino II) (97).

Traces of ochre concentrated on the skull or near the pelvic bones, as well as grave goods such as bone and stone knives, flints and pendants, have been found in both cemeteries (98). Faunal remains found in both cemeteries revealed some of the animal species that were exploited, which appeared typical of hunter-gatherers (elk, pine marten, water vole, wolf or dog, different fish species; and in the later stages: beaver, fox and bear). Although some burial goods exhibited parallels with Veret'e and Butovo archaeological cultures (97), the majority could be attributed to a period spanning from the second part of the Mesolithic to the Early Neolithic, which was confirmed by radiocarbon dating (99). Additional information came from bioarchaeological analyses of anthropological material from this site (39 individuals). The inspection of teeth revealed healthy conditions without any signs of caries. Enamel hypoplasia and calculus deposition were present at low levels, and showed the same intensity for both sexes. Further study of additional markers of physiological stress in children confirmed this finding, pointing to a similar status of male and female children in this hunter-gatherer society. Significant craniological variation was noted in the site. A series of male skulls from Minino exhibited clear analogies with synchronous populations of northeastern Europe as well as older Paleolithic hunters. In

early development stages of this region, the population exhibited high life expectancy, which decreased during later periods. Isotopic analyses also revealed a changing of diets over time (100). All these chronological dynamics of anthropological indicators could reflect the migration processes of the region.

Khvalynsk, Volga region, Russia (KVK001): KVK001 (museum nb. KO128) is a human skull fragment that was discovered in 1927 on the Khoroshensky Island of the Volga River near the town of Khvalynsk, by the local museum staff. It was found in a semi-submerged pebble layer, which also contained abundant remains of Quaternary fauna (101). Weinert (102), who first published the finding, attributed it to a human from the Paleolithic. Later, Gremyatsky (103) concluded that it belonged to a, possibly male, anatomically modern human with some Neanderthal features. As noted by Bader (104), the stratigraphy of alluvial deposits in the region is quite uniform and stable. After conducting additional analyzes, the researcher did not rule out that this skull fragment could be associated with the Upper Paleolithic (Aurignac-Solutré). While the pebble layer in which it was found, was attributed to the Mindel-Riss Interglacial, A. P. Ososkov (105) considered it possible that the sandstone load of this region was formed during the Late Paleolithic, and the results of radiocarbon dating indicated that the sample belongs to the Early Mesolithic time (OxA-23001, 9045 ±40 uncal BP).

“Tutkaul”, Tajikistan. (Individuals TTK002): The site of Tutkaul is located in southern Tajikistan, 70 km southeastwards of Dushanbe in the Dashti-Mazar region. The site was discovered during the archaeological survey of the Nurek dam’s flooding area led by A.P. Okladnikov in 1956 (106). Excavations were conducted during six field seasons (1963, 1965-1969) as part of a rescue archeological program led by the Tajik Archeological Expedition (107). The upper part of the stratigraphy consists of a medieval fortified settlement, followed by levels 1 and 2 which are attributed to the Hissar Neolithic culture. The lowest stratigraphic units (3 and 2a) belong to the Early and Late Epipaleolithic (108). At the base of level 2, three burials were identified containing the remains of four individuals: a female adult (burial nb. 1), a subadult (burial nb. 2) and two children (burial nb. 3). The burials were oriented to the SE-NW, the skeletons were in a bent condition on the left side, suggesting that the bodies were tied up before being buried. **TTK002** is the

vertebra from a child found in burial nb. 3. An unidentified bone fragment from the same burial was radiocarbon-dated to 8425-8025 cal BP (GV-02104 7450±106).

Grotta dell'Uzzo, Sicily, Italy (UZZ081, UZZ075, UZZ061 and UZZ099): Grotta dell'Uzzo is a large rockshelter on the eastern side of the San Vito lo Capo peninsula in NW Sicily. The discovery of the prehistoric deposits was made in the early 1970s by Giovanni Mannino (109), who excavated a trench exposing an in situ Mesolithic sequence. Later in the 1970s, 1980s and in 2004, within a number of trenches both inside and outside the overhang of the rockshelter, evidence was unearthed that occupation at the site occurred from the late Upper Palaeolithic through the Mesolithic and into the Neolithic (110–116). The cave was occupied in the early and middle Neolithic, as well as during the Bronze Age and throughout history, being used until recently by shepherds as a stable for sheep.

Grotta dell'Uzzo is a key site for Mediterranean prehistory because its long stratigraphic sequence covers the transition from hunter-gatherer to agro-pastoral economies (113, 114, 116–118). During the so-called Mesolithic-Neolithic transition, hunter-gatherer producers of Castelnovian-like lithic industries adapted to the climatic and environmental changes that occurred around the 8.2 kyr cal. BP event by exploiting large proportions of marine-based protein, mainly originating from the exploitation of cetaceans (119). This phase was followed by the introduction of elements of a farming-based economy by early Neolithic groups that arrived in NW Sicily around 8000 years ago (113, 119).

Another reason for which Grotta dell'Uzzo is noteworthy is that, during the Mesolithic, it was a burial site, where at least 13 individuals were inhumated in 11 burials (111, 120, 121). Human remains were, however, also found scattered in most of the deposits excavated both within and just outside of the overhang of the rockshelter. A thorough dating programme started by Mannino et al. (119) has demonstrated that the loose human skeletal remains date to all the main occupation phases of the cave, including the Mesolithic, Mesolithic-Neolithic transition and Neolithic.

The skeletal remains from Grotta dell'Uzzo that were sampled for this study are attributable to the Late Mesolithic and Neolithic phases of occupation. **UZZ081** is a temporal bone fragment recovered in the superficial layer of Trench U and directly radiocarbon dated to

8520-8180 cal. years BP (MAMS-40721: 7807±26 BP). This calibrated age is corrected for the marine reservoir effect, which in the case of this specimen is affected by around 45±10% marine protein ($\delta^{13}\text{C}$: -15.9‰; $\delta^{15}\text{N}$: 13.0‰), based on a calculation that takes into account the mixing model elaborated by Mannino et al. (119). **UZZ081** is thus contemporary to the so-called Mesolithic-Neolithic transition, which at Grotta dell'Uzzo was characterized by the presence of late Mesolithic hunter-gatherers who produced Castelnovian-like lithic industries (110).

UZZ075, **UZZ061** and **UZZ099** have all been recovered from deposits that post-date the Mesolithic-Neolithic transition, after which Grotta dell'Uzzo was occupied by farming and agro-pastoral communities. **UZZ075** is a temporal bone fragment recovered in Trench S (stratigraphic spit 5) and is directly dated to 7280-7160 cal. years BP (MAMS-40717: 6310±23 BP), which corresponds to the early Neolithic facies (Stentinello). **UZZ061** is a phalanx recovered from the superficial layer of Trench H and is directly dated to 6830-6660 cal. years BP (MAMS-48211: 5923±25 BP), which corresponds to a middle Neolithic facies called Stentinello / Trichrome / Serra d'Alto. All the dates reported here have been calibrated using OxCal 4.4 (122) and the most recent calibration curve IntCal20 (123). **UZZ099** is a temporal bone fragment which was radiocarbon dated to 5989-5907 cal BCE (MAMS-40714 5185 ±31).

Jazikovo (Yazykovo), Volga region, Russia (JAZ001): The remains of an adult woman (**JAZ001**, museum nb. 8619) were excavated at the Yazykovo peatland near the village of Yazykovo, Kashinsky district, Russia, in a cultural layer which was firstly discovered by B.S. Zhukov, A.E. Alikhova and M.V. Voevodsky in 1928 on the banks of the Yakhroma river. The researchers discovered ceramics, the stratigraphic position of which indicated that they were anterior to the Pit-Comb Ware culture (124). This pointed to Yazykovo being one of the most ancient Neolithic sites in the Upper Volga region. Today, Yazikovo encompasses three complexes (Yazikovo 1,2 and 3), which are attributed to the Central variant of the Upper-Volga archaeological culture (125).

Based on the analysis of ceramics from different sites including Lyalovo, Yazykovo 1 and 2, Nikolo-Perevoz 1 and 2, Kholomonikha, Lipki, Volosovo 1, Malo-Okulovskaya 1,

Malo-Borskaya and Balakhninskaya 3, Zhukov identified two variants characterizing the development of Neolithic societies in the Upper Volga region (124). The variant presumed to be the oldest one was defined as the “Yazykovo” type: thick-walled pottery with abundant grit in the dough, decorated with parallel carved stripes. The second variant was denominated as the “circular-pit comb complex”, due to similarities with the “comb” pottery and particular ornamental features. Further studies have led to a more detailed periodization of the Volga-Oka Neolithic, which divides it into an “early” and “developed” period. In the Upper Volga region, subsistence strategies were maintained across the Mesolithic to Neolithic transition, and the beginning of the Neolithic is solely marked by the appearance of pottery. The Early Neolithic period extends from 7100/7000 to 6100/6000 uncal BP (126), and is viewed as a period in which Yazykovo-type ceramics were imported in the region through occasional trade with non-autochthonous individuals, while the local lithic industry remained unchanged (127–129). The developed Neolithic period extends from 6100/6000 to 3800/3700 uncal BP (130), and is marked by the appearance of, likely locally produced, comb-ornamented ceramics, possibly reflecting population influx. **JAZ001** was radiocarbon dated to 6329 ±21 uncal BP (MAMS-37910), thus placing it at the end of the Early Neolithic period.

The Late Neolithic burial ground of Wenigensömmern, Thuringia, Germany (WSN010 and WSN012): The Late Neolithic (Bell Beaker Culture) burial ground of Wenigensömmern (Sömmerda County, Thuringia, Germany) comprises three small burial groups encompassing 14 graves and 18 individuals (131). The cemetery is located 2700 m southwest of the Early Bronze Age cemetery of Leubingen. It is dated to ca. 4300-4100 BP. The anthropological determination was made by Sabine Birkenbeil in 2016. **WSN010** and **WSN012** came from a large grave pit (feature 07/319-1129) that contained remains of four individuals, an undecorated bell beaker, and six flint artifacts. Only their skulls were found, secondarily deposited in the southeast corner of the burial pit, without any directly attributable grave goods. The other two individuals in the grave had been buried in crouched positions.

The Early Bronze Age cemetery of Leubingen, Thuringia, Germany (LEU065): The cemetery of the Early Bronze Age (Únětice Culture) of Leubingen (Sömmerda district,

Thuringia, Germany) is located about 750 m southwest of the large Leubingen Tumulus, a prosperous region at that time (*132*). However, the occupation of the burial ground predated (ca. 4100-3900 BP) the construction of the Tumulus in the classical phase of the Únětice Culture. It is the burial site of at least one settlement located in the direct vicinity. The completely investigated burial site includes 34 graves of the early to middle stages of the Únětice culture containing a total of 47 individuals. Other isolated graves were found in the vicinity. The cemetery is located 2700 m northeast of the Late Neolithic cemetery of Wenigensömmern. **LEU065** was first anthropologically determined by Bärbel Heußner in 2010, and comes from a double grave (feature 09/100-2045), a double burial, which took place with a clear time gap. It is an adult male whose remains laid disarticulated above a regularly buried mature female in a stone cist. No grave goods could be assigned to **LEU065**, while the female individual was buried with culturally specific grave goods: two vessels and a perforated shell disc.

Tell Yunatsite, Bulgaria (I0784 and YUN048): Tell Yunatsite is among the most prominent tells in Bulgaria (110x100x12m). It is situated in Southern Bulgaria, in the western part of the Upper Thracian Plain, 6 km northwest from the city of Pazardzhik, and approximately 1 km southwest of the modern village Yunatsite. The tell developed on a low terrace on the ancient bank of the Topolnitsa River, near to its confluence with the Maritsa River. It is located in a fertile plain bounded by mountains – the Rhodope Mountains to the south, Rila and Ihtimanska Sredna Gora Mountains to the west and Sashtinska Sredna Gora Mountain to the north. The first excavations of the site were carried out by V. Mikov in 1939. Systematic archaeological excavations of the tell's eastern section began in 1976 and continued until present (*133–136*). So far, the excavations have yielded evidence of occupation from the 5th millennium BCE until the 6th century CE – including Chalcolithic, Early Bronze Age, Iron Age and Roman Age occupations, as well as a Medieval cemetery. Long-term occupation was documented in two periods – the Chalcolithic and the Early Bronze Age.

Individuals **I0784** and **YUN048** were found in the latest Chalcolithic settlement (occupation level BI), which provided important information to understand the processes and events leading to the end of the Chalcolithic cultures of the Balkans. It was conquered,

burnt and the residents were slaughtered. Some of the bodies (ca. 50 individuals) were discovered buried under the remains of houses destroyed by fire. Some of the human skulls bear signs of injuries. They were probably inflicted by copper (?) battle axes and caused the death of the individuals (137). Bones of more than 20 individuals were found scattered between houses. **I0784** (grave 96) is a skeleton found in Building 5 from level BI, sq. K-J19, belonging to a 40-50 year-old woman (135, 137), and on which two skull traumas were identified. **YUN048** (grave sq. 9) was found in sq. O9, outside of Building 1 (to the north of it), at the “street level” of level BI. The death of this individual was also associated with the destruction of the last Chalcolithic settlement. Genetic sexing suggests that it belonged to a male.

Cueva de las Lechuzas, Spain (CLL005): This site is a cave located in the center of the Villena basin, on the eastern slope of the Cabezo de las Cuevas. The access of the cave was originally about 2 meters wide and gave way to the main room leaning towards the bottom, but it was then destroyed together with part of the cave during quarry exploitation. The archaeologist J.M. Soler identified human remains belonging to at least 18 individuals in the site, including **CLL005** (Arc. ID: Lech 7), an adult male. He also recovered an important set of grave goods consisting of several arrowheads, an axe and a hoe made of polished stone, several pendants made of various seashell species and teeth, and several necklace beads made of fish and stone vertebrae. Several bone punches and ceramic bowls were also found. These materials indicate a use of the cave during the Chalcolithic period (ca 5300-4300 BP).

Peñón de la Zorra, eastern cave, Spain (PLZ001): The cave is located in the eastern slope of the El Morrón mountain range at about 640 m asl. It was excavated by J.M. Soler in 1964. On the same hill, the village of Peñón de la Zorra was shown to have been occupied during several periods between the Bell-Beaker and the Bronze Age. Human remains from at least six individuals were recovered, including three adults and three children. Different objects were discovered with the human remains, including a dagger and a pair of Palmella-type points made of copper, a necklace made of fish vertebrae, a silver earring and small bowls. Direct dating of some individuals revealed the use of the cave during the Bronze Age, consistent with previous findings, although a previous use during the Bell-Beaker

period associated with copper weapons cannot be ruled out. Individual **PLZ001** (Arc. ID: Individuo 1) is an adult female burial which was indirectly dated to the Bronze Age (ca 4200-3500 BP).

Hostivice, Prague-West district, Czech Republic (HOP004): The site was discovered during rescue excavations led by J. Klementová and D. Daněček (Museum Roztoky) in 2007-2008 (*138*). The excavated area covered 10 ha, and more than 1,300 settlement pit features as well as a similar number of post holes were found. The site has been occupied during the Linear and Stroked Pottery cultures, the Funnel Beaker and Řivnáč cultures, Hallstatt (Ha C-D1), Roman Iron Age and Early Middle Ages. More than 33 graves were also uncovered and burials were identified in sunken settlement features from the Funnel Beaker (Baalberge stage), Corded Ware, Bell Beaker and Knovíz cultures (B D – Ha A2), as well as from the La Tène period (Lt B1-C). Anthropological analyses were performed by M. Dobisíková (National Museum). **HOP004** (Grave 22/Feature 691) is a skeleton found in a left-sided, crouched position, with the head towards the south, which was identified as a 40 to 60-year-old female. No grave goods were found with the individual. The individual was dated to the Bell Beaker period based on the archaeological context, and radiocarbon dated to 4400–4100 cal BP (MAMS-38921: 3826±27).

Tell Atchana (Alalakh), Turkey (ALA098 and ALA110): The site is located in the Amuq Valley, Turkey, and belongs geographically to the northernmost stretches of the Levant. It was founded in the terminal Early Bronze Age or earliest Middle Bronze Age (ca. 2200-2000 BCE) and quickly developed into an urban site with palaces, temples, workshop areas and fortification systems during the Middle Bronze Age II (from ca. 1800 BCE on). Textual evidence attests to its significance as the capital of a regional kingdom during the Middle and Late Bronze Age (ca. 1800-1300 BCE), until its nearly complete abandonment around ca. 1300 BCE (*139–146*). Excavations at Tell Atchana were undertaken by Sir Leonard Woolley from 1935-1939 and from 1946-1949 (*147, 148*). New excavations under direction of K. Aslihan Yener have taken place since 2003 (*144, 145*).

Individuals **ALA098** and **ALA110** stem from these new excavations. They were found in the so-called extramural cemetery in Area 3, located right outside the city wall, in which a

total of 134 burials were excavated dating to the Middle and Late Bronze Age (149–151). Most of them were laid out in simple pits oriented northwest-southeast parallel to the city wall and follow the outline of the slope. The majority of the graves are single burials (as is generally the case in Alalakh) (151). Most individuals were buried on their backs or right or left side with their legs flexed on one side (~80%) (149). Additionally, eight secondary burials were identified in the extramural cemetery, four of which only consist of two or three skulls and/or mandibles buried together (149). Compared to the intramural burials (on average 3.49 objects per individual), especially those in the Royal Precinct, the burials in the extramural cemetery contain fewer (on average only 1.25 per individual) and less varied grave goods. 58.2% of the graves here did not contain any grave goods, and the others were typically furnished with one to two objects, mainly consisting of pottery and jewelry (149, 151).

Individual **ALA098** (square 45.45, locus 23), was found in a secondary burial of three mandibles, together with a short-neck jar. The sex and age of this individual are unknown. Strontium isotope analyses have shown that this individual likely grew up outside the Amuq Valley (149). Individual **ALA110** (square 45.45, locus 48) was found in a single primary burial containing no grave goods. The individual was identified as a 65 to 75-year-old male, and strontium isotope analyses suggested that this individual moved to Alalakh after the formation of the M2, but before the M3 was formed (149).

The Middle Bronze Age burials from the tell of Kamid el-Loz, Lebanon (KEL045):

The tell of Kamid el-Loz is located in the Beqa'a-plain in Lebanon and is surrounded by the present-day village of the same name (152). Today, a cemetery covers the southeastern parts of the tell. Excavations on the tell took place from 1963-1981. They were first conducted by a team from the Universität des Saarlandes (Germany) led by Rolf Hachmann (153, 154) and were renewed in 1997 by Marlies Heinz from the Albert-Ludwigs-Universität Freiburg (Germany) (152, 155). The earliest traces of settlement uncovered date to the 3rd millennium BCE (Early Bronze Age) (155). During the Middle Bronze II (ca. 1750-1550 BCE) Kamid el-Loz developed from a village into a city with a palace, a temple, and other administrative buildings, as well as a fortification system and regular living quarters. After the collapse of this first city, inhabitants of either the tell or the

surrounding valley used the ruins of two buildings on the northern slopes (“Gebäude I and II”) as a cemetery (152), from which 24 individuals (7 adults and 17 children) were excavated. The graves were dug directly into the ruins of the buildings, often re-using older walls as grave linings and stones and mud bricks as covering. 11 individuals were buried in simple pits in primary position, four graves were single jar burials of small infants, and nine individuals were buried in groups of three which may represent multiple burials or accidental use of older burials. The positioning of the body was recognizable in 19 cases: 14 Individuals were buried on either right or left side with legs bent (“Hocker”) and 5 individuals were buried extended on their back. The deceased were accompanied by none or only a few grave goods: nine graves were lacking any burial goods, including the four pot burials. The other 15 graves contained pottery vessels, jewelry/personal adornments and/or animal bones and one bronze hatchet in grave 109 (156–158).

Individual **KEL045** (grave 97, Areal ID15, ID15:3, control-list number KL 67:411a): is a female individual of minimum 60 years at death (anthropological age and sex estimate performed by Edith Oplesch; sex estimate confirmed by aDNA-analysis). She was buried in an elongated oval, stone lined pit that was built into a WSW-ENE running wall of former building I (“Gebäude I”). The pelvis, right femur and chest area of the skeleton were strongly disturbed. The pelvis was separated from the upper body and lying 20 cm higher. The skull was fragmented and the mandible slid a bit to the side. The skeleton was oriented ENE-WSW, with the head in ENE. The deceased was buried lying on her left body side with legs flexed (“Hocker”). The left arm was extended and running diagonally from the shoulder to the knees so that the left hand was positioned between the legs. The right arm was strongly bent and the right hand resting in front of the face. A ceramic bowl was lying in front of the face in the SE corner of the grave, a ceramic juglet was placed at the right elbow. A broken bronze pin was lying in the chest area (157). A tooth of this individual was radiocarbon dated to 1886-1754 cal BCE (MAMS-43549, 3498±21).

Karanaevsky ground burial, Republic of Bashkortostan, Russia (KAP002): The Karanaevsky ground burial is located 0.4 km northeast of the northern outskirts of the village of Karanaevo, on a high flat plateau on the right bank of the river Kushkain in Sterlibashevsky district (Republic of Bashkortostan, Russia). It consists of 13 mounds with

diameters of 10 to 87 m, and heights of 0.2 to 1.5 m. Kurgan nb. 7 (12 m in diameter, about 0.43 m high) was excavated in 2008. It contained 6 burials. An adult man was buried in burial nb. 3, a woman in burial nb. 4, while children and adolescents were buried in others. The funeral rites and ceramics of the site are characteristic of the Srubnaya culture tribes of the Southern Urals. Burial nb. 5 contained an adolescent and ~5-year-old child (**KAP002**) buried together, together with two ceramic vessels.

“Ipogeo degli Avari”, Trinitapoli, Apulia, Italy. (TRI011): The Bronze Age Madonna di Loreto site (Trinitapoli) is located in the Tavoliere delle Puglie, a plain in northern Apulia, southern Italy. The site comprises several hypogeal structures intended for cultic and, in a later phase, funerary function. The hypogea are artificial underground chambers excavated in calcareous banks and provided with a narrow and steep open-air entryway (dromos) followed by a low underground passage (stomion). At present, seven monumental hypogea are known (*159*), among which the “Bronzes Hypogeum,” the “Fermatreccia Hypogeum,” the “Guardian Hypogeum” and the “Ivories Hypogeum”, from which comes the sample **TRI011**. The Ivories Hypogeum was excavated between 1999 and 2001 by A.M. Tunzi under the former Soprintendenza per i Beni Archeologici della Puglia. As for the other structures, the hypogeum was probably first built and used for fertility cults and ceremonies. After a phase of abandonment, during the last phase of the Middle Bronze Age, the structure underwent some structural modifications and was transformed into an elite grave (*160*).

The archaeological and bioarchaeological investigation allowed the reconstruction of the depositions’ modality. In two distinct phases, dozens of corpses were progressively placed in the stomion and chamber, along with several cultural items: bronze objects, amber beads, vases, pierced shells, and two small ivory sculptures, hence the site’s name (*161, 162*). As new individuals were added, those below were displaced with a partial or total loss of anatomical connections. Thus, the skeletal series is mainly represented by commingled bones of more than 100 individuals in different preservation states. In a few cases, based on spatial proximity and biological characteristics, it was possible to assign bones to specific individuals. The preliminary anthropological analysis shows the presence of

infants, children, and adults of both sexes. **TRI011** is a deciduous mandibular second molar of a 4-6 years child coming from the stomion.

Iron Age burials in Latsch/Vinschgau, South Tyrol, Italy (I0216, I0217): In 2007, several Iron Age burials were discovered during construction work in Latsch (Vinschgau). Archaeological investigations by the Amt für Bodendenkmäler, Bozen (Office for Archaeological Monuments, Bolzano) showed that, in this area, the Adige river had been dammed back several times as a result of the activity of the Tarsch debris-flow cone, creating a lake and marsh landscape. Several large stone blocks on the northern edge of the bank, broken off from the steeply rising rock face, formed the reference point for inhumation burials of the Iron Age. The bone material recovered during archaeological investigations indicates a minimum number of 14 individuals, including 10 adults and 4 children. The burials contained remarkably little jewelry and grave goods: a single amber bead, finger rings, small rings worn in plaits, a bronze bracelet, part of a ceramic vessel and animal bones (sheep/goat, pig, cattle, dog). In addition to typological dating of the findings, which placed the burials in the Hallstatt and Early La Tène periods, a series of ¹⁴C dating was carried out: 800 - 480 cal BCE (91.1%); 780 - 480 cal BCE (87.3%); 770 - 410 cal BCE (95%); 600 - 390 cal BCE (71.3%); 400 - 160 cal BCE (95.4%).

Burial practices appeared to depend on the available space. Stone linings or covers were not observed. One woman was probably buried in a crouched position (Hocker) in a niche, while another individual was buried laterally in a crouched position (Hocker) with arms bent. At least two individuals, an adult woman and a child were buried together with their legs intertwined. Some of the skeletons were completely surrounded by river sediments, indicating that the river had risen at least temporarily above the burial level, altering the original body positions. It is remarkable that the deceased were not buried according to the local cremation practices of the Fritzens-Sanzeno culture typical for South Tyrol, North Tyrol and Trentino, which suggests that these individuals were part of a group of settlers who had migrated to the Vinschgau Valley and retained the burial customs of their place of origin.

Chotín, Komárno district, Slovakia (CHT001): At Chotín, two Celtic cemeteries were found. The first La Tène (LT) cemetery was found southeast of the village (site VIII) during rescue excavations led in 1960-61 by M. Dušek (IA SAS). 16 burials were initially discovered, but more graves were revealed and destroyed east of the excavated area as a result of sand extraction conducted in 1965, three of which were inventoried by Priska Ratimorská. In 1975, investigations led by J. Bujna led to the discovery of four additional graves, one of which contained burial goods. The graves date from the end of the LT B1 to the LT B2 periods, *i.e.* to the last quarter of the 4th to the first half of the 3rd century BCE.

Another biritual LT cemetery was found west of the village, at “Chotín X”. Initial graves were discovered and destroyed during sand extraction, but the inventory could be saved for three of them (nb. 1-3). During rescue excavations led in 1971-72 (Podunajské Múzeum Komárno, under the direction of P. Ratimorská), 44 additional graves were excavated (163). Among the total of 47 graves discovered, only four are cremation graves. Five graves were weapon-equipped tombs, and an iron sword in a scabbard was found in all of them. In a warrior’s grave (nb. 14), a two-piece bronze arm bracelet with a hinged closure and rich filigree decoration was found. The grave of a rich woman (nb. 21) several pottery items were found, including a vessel with a profiled foot of Hellenistic style (a kantharos). In grave nb. 34, a man was buried with a gold and a silver ring on the left hand as well a set of tools comprising an axe and three files (164). Further analyses of the skeletons and as well as genetic investigation have been conducted on this site (164–168). Individual CHT001 was a 23 to 32-year-old male which was buried in grave 12 (excavated in 1971) at the site Chotín “X”. It was found with two bronze and two iron made fibulae close to the upper part of the body, a bronze (?) bracelet at the right upper arm, two iron “Koppelringe” around the pelvic region, two bronze hollow casted buckle anklets at the feet as well as a ceramic vessel.

Akbeit burial ground, Kazakhstan (AKB003): The large Akbeit burial ground belongs to the Tasmola culture, which developed in the steppes of Central Kazakhstan during the 8th-5th centuries BCE (169). Studies have highlighted connections between the Tasmola population and tribes of eastern regions of Central Asia (170), showing that early Saka cultures were much closer to each other than previously thought. Individual **AKB003** was

found in a grave in Kurgan 7, which is located in the first of the four groups of the Akbeit burial ground, and has a diameter of 21 m and a height of 2.5 m. The grave had the following dimensions: length 1.9 m, width 1.8 m, and depth 0.95 m. The skeleton, belonging to a 4 to 5-year-old child, was found at the bottom of the grave. Objects found with the skeleton such as a gold necklace, two gold earrings with turquoise inserts, a bronze mirror decorated with gold, and a small bone case likely containing a red cosmetic substance, suggested that the individual was a female belonging to the elite of the Tasmola society (171, 172).

Berel necropolis, Kazakhstan (BRE008, BRE026 and BRE028): The Berel necropolis was discovered around elite kurgans of the Pazyryk time, and has later revealed materials identified as belonging to the Hun-Xianbei cultural-chronological horizon (1st-4th century CE), which fits into the broad framework of the so-called Great Migration of Peoples era (2nd century BCE - 6th century CE) (173). To date, it encompasses over 50 mounds of different sizes, half of which are either cenotaphs or altar, and the others containing simple shallow burials in soil pits, stone boxes or wooden structures. These various burial structures, together with different orientations of the deaths, and the occasional inclusion of horses in the burials, point to ethnocultural diversity. In particular, some of the materials from the Hun-Xianbei period found in the site exhibit direct analogies with the Transbaikal region.

Individual **BRE008** (Berel 2017_90A) was excavated from a ~3 m diameter mound, surmounted by an irregular circle of stones. The skeleton was found at a depth of 60 cm, in an elongated position on the back, with the head oriented to the west. The right clavicle and bones of the right hand appeared to have been moved, most likely by rodents. The left foot was also missing. The only objects found with the skeleton were sheep bones and a corroded iron object (probably a meat knife). The individual was identified as a female and radiocarbon-dated to 258-366 cal AD (MAMS-42126 1730 ±13). Individual **BRE026** (Berel 13) was found in burial nb. 13, in which the remains of two horses were also discovered. The burial also contained a set of weapons characteristic of a lightly armed equestrian warrior from the Hun-Xianbei period of the Early Medieval Kazakh Altai (173). Individual **BRE028** (Berel 2) was excavated from an elite burial mound from the Pazyryk

period, of ~28 m diameter and 0.7 m height (mound nb. 2). It contained a large chamber (4.9x4.0 m, more than 4 m depth), which itself contained a wooden funerary cabin in which the remains of the individual were found. Several objects were also found in the burial, including clothes decorated with gold foil, a bronze mirror with a zoomorphic handle, a grater made of light gray fine-grained granite, grains of cereals, ceramic vessels, a wooden table, a horse tail and an iron knife. The funerary chamber also contained the remains of seven horses found with golden ornaments depicting fantastic polymorphic creatures near the head. These elements are characteristic of the 4th-3rd c. BCE in Berel.

Kuelap archaeological complex, Chachapoyas, Amazonas, Peru (KUE033): The Kuelap archaeological complex belongs to the Chachapoya culture, and is located in the Tingo district, Luya province, Amazonas region, at 3000 m asl. The Chachapoya culture flourished around 800 CE until conquered by the Inca and the Spanish. Kuelap, located at 3000 m asl on the left bank of the Utcubamba river, is one of the largest sites of this kind (~450 ha), which is composed of public and residential buildings and also includes funerary areas and agricultural terraces as well as massive perimeter walls. The perimeter walls enclose various differentiated areas with about 420 structures, most of which exhibit a circular layout. The functionality of the site is still debated. Due to several architectural features, its location and surroundings, it has been suggested as a fortified city, a residence of a centralized elite and even a center of astronomical importance (174). There have been various seasons of archaeological work in different areas of the site since the 1980's and until now, which have recovered hundreds of skeletonized human remains. **KUE033** comes from the "La Fortaleza" area (South Sector, Structure 3, E3, burial 12A) and was excavated in 1989. The preliminary bioanthropological assessment made during sample collection suggested it is an adult male.

Purunllacta de Soloco, Chachapoyas, Amazonas, Peru (PLS004): The site is located in the Sonche valley at an altitude of around 3000 m asl. Purunllacta de Soloco and other sites in this region show cultural features (pottery and architecture) that indicate Chachapoya or Late Intermediate (1000-1475 CE) and Late Horizon or Inca (1475-1532 CE) occupations; however, the site also has evidence of occupation during the colonial period following the Spanish conquest. This site and others situated on the right bank of the Utcubamba river

were integrated to the Qhapaq Ñan Inca road system to a certain degree. In addition to buildings with a circular layout, typical of Chachapoya settlements, there are structures in “D” shape, which have also been found elsewhere in the region, e.g. Kuelap. This architectural style seems to have been frequent during the Late Horizon and their function as ceremonial structures has been suggested. The site has two sectors, the first (Sector A) at a lower elevation and associated with the Inca road, and the second (Sector B) at a higher elevation (ridgetop) (175). The site was excavated in 2014 and 2015 and researchers recovered artifacts and human remains associated with the Inca and colonial periods. **PLS004** was recovered from Sector A, unit 14 (Nivel 3, Rasgo 4, burial 7). The preliminary bioanthropological assessment made during sample collection suggested it is an adult female.

The Malaya Ryazan II settlement, Samara region, Russia (MLR005): This site is located on the right bank of the Volga river in the southern part of Samarskaya Luka (Samara region). It extends over about 30 hectares. The cultural layer of the Middle Ages dates to the 13th-14th century CE. The settlement was a trade and craft village of the Ulus Jochi (Golden Horde), in which a predominantly Russian population lived (176). A cemetery was located in the eastern part of the village. In 2007-2012, 131 burials were excavated. All adult burials were males, with the exception of 2 females, and 63 burials were children. **MLR005** (Burial 15) was a ca. 55-year-old male, laid on his back. The lower jaw was displaced, and some bones were partially destroyed.

Mayachny Bugor ground burial, Astrakhan region, Russia (MAY017): The ground burial “Mayachny Bugor” is located 400 meters north of Krasny Yar village outskirts (Astrakhan region, Russia), on the eponymous Baer knoll, and is conventionally divided into 4 parts: Mayachny Bugor-I, II, III and IV. **MAY017** was found in burial 80, which belongs to Mayachny Bugor-II, on the northeastern part of the knoll, and is a burial pit with an irregular trapezoidal shape and a rounded eastern wall. (~ 116x30 cm). Wooden remains were found in the pit. The individual was a 5.5-6-year-old child, laid on his back, with the head directed to the west-northwest. The skull was crushed by the soil, but it could be determined that it was laid face up and slightly inclined to the right. The left half of the skeleton was destroyed, likely by animals, with the bones shifted from their original

position. The right arm was extended closely along the body, with the wrist close to the right-wing of the pelvis. The legs were stretched out. The bones showed signs of iron deficiency anaemia. This burial belongs to the Golden Horde period (13th-14th century CE).

Ancient HBV genome reconstruction

The number of reads screened per library prior to HBV DNA enrichment ranged from 1.3M to 1,485M (median=5.1M). The fraction of HBV-mapping reads in positive individuals ranged from 0.01 to 77.95 reads/M (median=0.78/M). After targeted DNA enrichment, the fraction of HBV-mapping reads increased by several orders of magnitude (median: 47,377-fold), which allowed us to reconstruct a significant fraction of the HBV genome in most cases. On average, 72% of the genome was covered at least 3x, and that proportion exceeded 90% in 66 individuals from which almost complete genomes could be reconstructed (fig. S9A,B, Data S1).

Typical patterns of ancient DNA damage were observed in most cases, with an average C to T substitution rate at the 5' read end of 19.3% and 6.4% for non-UDG and half-UDG treated libraries, respectively (fig. S9E; Data S3). The frequency of damage-like substitution observed on HBV-mapping reads correlated well with that observed on human-mapping reads across libraries (R-squared: 0.71; fig. S9F). When only a low number of HBV-mapping reads were recovered, or when a full UDG treatment was employed for library preparation, it was not possible to properly estimate DNA damage. However, sample contamination with HBV DNA is unexpected, and the phylogenetic placement of all reconstructed HBV genomes was consistent with their spatio-temporal origins, supporting their authenticity (fig. S3; table S2).

Teeth are generally considered as the best DNA reservoir for ancient bloodborne pathogens (177, 178). DNA libraries prepared from teeth indeed yielded significantly higher HBV genome coverage than those prepared from bones (ratio of estimated marginal means: 1.9, p-value: 0.0071; fig. S9C). However, it is remarkable that complete HBV genomes could be recovered from a variety of skeletal elements, including from the low vascularized

petrous bone. Additionally, no significant difference of coverage was detected between libraries prepared from petrous and other bones (p-value: 0.99).

The HBV genome is partially double-stranded, with a single-stranded portion of variable length covering about one fourth of the genome (69). On average, coverage was ~6 times higher in the double stranded portion of the genome compared to the single-stranded region (fig. S9B). However, a significant proportion of the single-stranded region could be reconstructed in most cases, even when a double-stranded protocol was used for library preparation. This is consistent with variation in length of the single-stranded region among circulating viral particles, some of which can exhibit completely double-stranded genomes (69). Using a single-stranded instead of a double-stranded protocol for library preparation led to an increase of HBV genome coverage (ratio of estimated marginal means: 6.2, p-value: $3e-4$; fig. S9D). On the other hand, the use of a single-stranded library protocol did not lead to a significant reduction of the coverage ratio between double and single-stranded regions (p-value: 0.061). This might, however, be due to a lack of detection power, since only 8 out of 154 libraries were prepared using a single-stranded protocol in our study.

Phylogenetic analyses

Model selection strongly supported the use of a relaxed clock instead of a strict clock model, and the use of a coalescent skyline population model with 5 time intervals was favored (table S5). Bayesian estimation of temporal signal showed overwhelming support for the model incorporating tip dates (log Bayes factor: 338), indicating a strong molecular clock signal in our dataset, consistent with the results of root-to-tip regression (fig. S3). We estimate an HBV substitution rate of $1.5e-05$ subst.site-1.year-1 (95%HPD: $1.3e-05$ - $1.7e-05$) and a tMRCA of 13,518 BP (95%HPD: 12119-15680). These estimates were relatively robust with respect to different aspects of the dataset (table S1). On the other hand, the use of a time-dependent rate (TDR) model yielded a significantly higher tMRCA (95% HPD: 17170-20240).

Our estimated maximum clade credibility (MCC) tree exhibits the most frequently reported topology so far, in which the oldest split separates the American genotypes from those recovered in the rest of the world (18) (which we refer to as the Eurasian branch). The

monophyly of the American HBV branch, including all ancient American strains dating back to as far as ~9 ka, was highly supported (Fig. 2). Ancient strains from western Eurasia were either grouped with modern genotypes A or D, or formed distinct clades that appeared consistent with their spatiotemporal distribution: Mesolithic clades 1 and 2, Anatolian Early Neolithic branch (composed of a single genome) and the Western Eurasian Neolithic-to-Bronze Age clade (WENBA, from which genotype G appears to descend). The respective monophyly of all these lineages, as well as that of the other modern genotypes, was highly supported (Fig. 2). Conversely, deep branching patterns were not well resolved. In particular, the monophyly of the Eurasian HBV branch was not well supported, and the exploration of plausible alternative topologies pointed to the possibility that some of the Eurasian lineages had diverged before the American branch (fig. S4). The placement of modern Eurasian genotypes relative to ancient lineages also appeared uncertain: while genotypes A, B, C, D, E and I formed a clade branching off between the Mesolithic 1 and later ancient lineages when using the full dataset, they appeared as a sister group to all ancient lineages when excluding modern data or genomes recovered from individuals showing signs of mixed infections, and the placement of this clade was not well supported in any analysis (fig. S5). Furthermore, the clade formed by genotypes D and E was recovered within the WENBA clade after exclusion of identified recombining regions (fig. S5D), which suggests that the polymerase region of these genotypes descends from a WENBA strain. Finally, the placement of genotype A relative to genotypes B and C also appeared poorly resolved.

Although these phylogenetic ambiguities do not question the main conclusions of this study, they highlight that further work is needed to decipher the early phylogeographic history of HBV and to decipher the evolutionary origin of the main genotypes prevailing today. Of note, the use of a TDR model yielded 100% support for the monophyly of the Eurasian branch, and a dating for the split of the Eurasian and American branch that was more in line with current estimates for the genetic divergence of ancestral First Americans (fig. S5B). Therefore, while model selection favored the use of a relaxed clock model (log BF: 405), the TDR model appeared to yield more consistent deep branching patterns. Furthermore, the TDR was selected over a strict clock model (log BF: 29). Thus, models accounting for both time-dependency and inter-lineage variation of rates would certainly

be useful for further investigation. Finally, the use of phylogenetic methods explicitly accounting for genetic recombination, as well as the reconstruction of additional ancient HBV genomes from more eastern parts of Eurasia may also provide better resolution regarding these questions in the future.

Genotyping of ancient HBV strains

We used the *mPTP* species delimitation tool (179) to classify ancient HBV strains into genotypes, as previously described (180). The method grouped all modern genomes in accordance with conventionally defined genotypes, with two exceptions: (i) genotype C and I were grouped together, consistent with previous observations that the genetic distance between some strains of these genotypes is below the usual delimitation threshold (3), (ii) genotype J was grouped with lineages found in gibbons and orangutans.

All ancient strains recovered on the branch leading to genotype A and D were classified as such, except for RISE387, which was assigned its own genotype. The clade formed by KVK001 and younger Mesolithic genomes from Europe as well as the one recovered from an early farmer of Anatolia (BON020) were classified into two respective genotypes, while the two oldest Mesolithic genomes (MPR001 and MN2003) were grouped with Southeast Asian NHP lineages and genotype J. The WENBA clade was classified into ten different genotypes, with genotype G identified as a separate genotype. The two most ancient American strains (CUN002 and CAO009) were assigned their own respective genotypes, while two strains from Peru dating back to ~500 years ago. were grouped together. Two other strains from the same period found in Peru and Mexico were grouped with genotypes F and H respectively.

The examination of interstrain genetic distances highlighted difficulties in reconciling conventionally used genotype delimitation thresholds ($>7.5\%$ intergroup divergence across the complete genome (3)) with a time-continuous phylogenetic framework (table S4). Indeed, ancient genomes may represent intermediate evolutionary steps, blurring frontiers between genetic groups that appear consistent when looking at modern diversity alone. While ancient strains classified as genotype A or D by *mPTP* indeed showed less than 7.5% divergence with any modern strain of the respective genotype (except for RISE387), they

also showed <7.5% divergence with ancient and modern strains from other genotypes. In general, grouping ancient strains based on the standard approach would have resulted in the delimitation of genotypes that did not form monophyletic lineages. For instance, the genetic distance between any European Mesolithic or Anatolian Early Neolithic strains didn't exceed 5.3%, but those formed three separate lineages. While the WENBA clade (excluding genotype G) was classified into 9 genotypes by *mPTP*, the maximum interstrain divergence in that entire clade was 8.6%.

While phylogeny-aware tools such as *mPTP* can represent interesting classification alternatives in this context, they are sensitive to phylogenetic uncertainties. For example, some of the WENBA subclades classified as genotypes were poorly supported, and not recovered as monophyletic in our time-calibrated phylogeny. Therefore, we used a provisional classification scheme for ancient strains based on phylogenetic clades that appeared robust to phylogenetic uncertainty and reflected a consistent spatiotemporal distribution: Mesolithic lineages 1 and 2, Anatolian Early Neolithic lineage, Western-Eurasian Neolithic-to-Bronze-Age (WENBA) lineage, Ancient American lineages 1,2 and 3 (fig. S2). Ancient strains grouping monophyletically with a specific modern genotype were classified as such. Using *EPA-ng*, we were able to assign modern genotypes or previously defined ancient HBV lineages to all low-coverage genomes except one (table S2; fig. S3). Likelihood weight ratios were close to 1 in the vast majority of cases, and all assigned lineages appeared consistent with the spatio-temporal context of corresponding individuals. The only genome which couldn't be assigned to a lineage was recovered from an individual from Tajikistan dated to ~8.2 ka, which appeared to belong to an ancient lineage that is not represented in our phylogeny (fig. S3).

Recombination analysis

Homologous genetic recombination is known to occur in HBV, and it is largely accepted that entire HBV genotypes descend from genomic hybrids (181). An exploratory search performed using *RDP4* (84) revealed signals for 30 putative recombination events in our dataset. Further investigation of the results allowed to retain 5 well supported events (fig. S10). These included previously reported findings (181–183): (i) we identified

genotype E as descending from a recombinant between an unknown lineage and genotype D, (ii) we identified the clade formed by subgenotypes B2, B3 and B4 as descending from a recombinant between subgenotype B1 and genotype C, (iii) we identified genotype I as descending from a recombinant between genotype C and an unknown lineage, and (iv), we identified a genome isolated from a wild chimpanzee in East Africa (AN: AF498266) as a recombinant between a typical chimpanzee strain and human genotype C. For all these recombination events, except the last one, identified recombination breakpoints were around the edge of the polymerase gene.

Previous analyses of ancient HBV genomes suggested that genotype A originated from a recombination event between an unknown lineage and genotype D around the polymerase gene (14). We recovered a well-supported signal for a recombination event involving genotypes A and D in the same genomic region, which was recovered in all ancient and modern genomes belonging to these respective genotypes. However, although genotype A was marked as the recombinant in the initial exploratory search, possible misidentification of recombinant and parental genomes was indicated. Examination of local ML trees rather pointed to genotype D being the recombinant, since it appeared as a monophyletic clade within genotype A in the tree corresponding to the recombining portion of the genome. Additionally, a phylogenetic analysis excluding the identified recombining region in genotype D suggests that its second parental strain belonged to the WENBA clade (fig. S5d). However, these results must be taken with caution, since the reconstruction of local phylogenetic trees could suffer from biases related to other recombination events, as well as from the loss of overall phylogenetic signal.

Our results are in accordance with the established view that the genetic diversity of HBV was largely shaped by genetic recombination, and that several of the main HBV genotypes prevailing today descend from ancestral recombinants. We employed a conservative approach to identify a set of well supported recombination events. However, the latter cannot be regarded as exhaustive. In particular, the initial exploratory search suggested several putative recombination events involving genomes from the WENBA clade. However, none of these was supported by all methods or could be independently reconciled with a consistent evolutionary scenario based on local phylogenetic topologies. This

highlights the difficulty to disentangle the evolutionary history of this virus which likely encompasses multiple and successive recombination events affecting overlapping regions of the genome. Ancient HBV genomes contain precious information in this regard, since they may belong to lineages involved in past recombination events that are more difficult to detect after thousands of years of evolution. Nevertheless, recombination analyses such as the one conducted here are unable to provide comprehensive phylogenetic pictures. Conversely, although classical phylogenetic methods are useful for describing and dating the main evolutionary divergences of this virus, they cannot capture the full reticulated process. In this regard, the use of phylogenetic models explicitly accounting for genetic recombination should be helpful in the future.

Genomic characteristics of ancient HBV strains

Some of the ancient strains from the WENBA clade carried an insertion close to the 5' end of the C gene associated with a stop codon in the pre-core region, similarly to genotype G. This was exclusively observed in 14 strains forming the subclade from which genotype G descends in our phylogeny. The insertion carried by most of these ancient strains was 39-nt-long and appeared non-homologous to the insertion found in genotype G in part of its sequence (fig. S8). However, in the two Bronze Age genomes which appeared the most closely related to genotype G (SGR003 and VLI060), the insertion was 36-nt-long and strictly identical to the one found in genotype G.

The 6-nt insertion typically found in the core gene of modern genotype A strains was recovered in several ancient strains belonging to genotype A, dating back to as far as ~2 ka. (SJN001, Abusi1543, DA119 and I1321), but not in earlier strains from that lineage. This points to the acquisition of that insertion shortly before the divergence time of all modern genotype A subtypes, which we date to around 3 ka.

The length of the pre-S1 region of the HBV genome varies across genotypes due to the presence of large indels near the 5' end of this reading frame (3). The shortest pre-S1 region is found in genotype D as well as non-human primate genotypes, and has a typical length of 324 nucleotides. This was also the case for all the HBV genomes dating back to between ~11 and ~7.5 ka. (i.e. the oldest recovered in this study), as well as many genomes in the

WENBA clade (fig. S8). This suggests that the MRCA of all HBV lineages carried this form of the pre-S1 region, and that insertions must have occurred later on, contrasting with the classical perspective in which genotype D and NHP strains exhibit a 33-nt deletion in this region (3). Instead, it appears more phylogenetically consistent to say that genotypes A, B, C and I carry a 33-nt insertion (also found in all ancient relatives of genotype A) which must have existed in their most recent common ancestor. Genotype F and H exhibit another 33-nt insertion at the same genome location, which we recovered in all ancient American genomes except the oldest one (CUN002). Other types of pre-S1 insertions, located near the 5' end of the reading frame, were found among WENBA HBV strains: a 12-nt insertion found in 33 strains, a 30-nt insertion found in four of them (MK5004, MK5009, LEU065 and PDA003), and a 30-nt insertion found in genotype G and its closest ancient relatives (SGR003 and VLI060). Additionally, 42 genomes from the WENBA clade contained a 36-nt insertion around position 224 of the pre-S1 region (using genotype D numbering), and the oldest American strain (CUN002) carried a 12-nt insertion at the same position.

Most of the above-mentioned insertions were observed in sets of genomes forming monophyletic groups in the phylogeny, further supporting the reconstructed topology (of note, all regions containing insertions were masked from the alignment used for phylogenetic analyses). However, two of them were found to be shared between distant lineages, suggesting ancestral recombination events that were not detected during the recombination analysis: three genomes from the WENBA clade (KAP002, MIB011 and RISE154) carried the 33-nt insertion found genotypes A,B,C and I, and the 30-nt insertion found in genotype G was also found in genotype E.

Origin of HBV in non-human apes

Non-human hominoids have also been found infected by HBV, with two specific HBV lineages infecting African (chimpanzees and gorillas) and Southeast Asian apes (gibbons and orangutans) respectively. Most HBV phylogenies recover these non-human-primate (NHP) HBV lineages at interspersed positions between human-specific genotypes, which has been thought to reflect separate spillover events from humans to wild primates in the

past (7, 28). Our phylogenetic reconstruction and dating also supports this idea since NHP lineages are recovered as two separate clades branching within the Eurasian branch (Fig. 2). An alternative scenario involving two HBV lineages coevolving with NHPs and being transmitted independently to humans seems incompatible with the timescale of the HBV phylogeny, which we estimate to be ~ 3 orders of magnitude shorter than that of hominoids (184). Other intermediate hypotheses involving several NHP-to-human and human-to-NHP transmission events are possible but appear less parsimonious given currently available data.

Southeast Asian and African NHP lineages are recovered as a sister clade to the Mesolithic clades 1 and 2 (although with low support in the first case), with median estimates of divergence times around ~ 11 and ~ 12 ka, respectively. This may indicate that HBV was transmitted to NHP by hunter gatherers around that period. HBV transmission from humans to wild NHP might have involved contact with infected body fluids during aggressive interactions (185, 186), which could indeed be more likely to occur in human populations relying largely on hunting for subsistence (187, 188). However, these divergence times represent earliest estimates for the date of these potential human-to-NHP spillover events (189). Physical interactions between humans and NHPs in these regions have likely continuously occurred and still happen today (190–192). Additionally, given that the distribution of NHPs has likely not changed dramatically during the past ten thousand years, these putative spillover events should have happened in African and Southeast Asian rainforests respectively. Since NHP HBV lineages branch off from lineages recovered in Europe in our phylogenetic reconstruction, a significant part of the biogeographic history leading to the spread of the virus into these Asian and African apes remains to be revealed. Furthermore, it has been shown that accounting for host-specific differences in viral evolutionary rates may be important for robust phylogenetic reconstruction (193). Thus, in the future, the retrieval of HBV genomes from ancient NHP specimens combined with the use of host-specific clock models could be helpful to further assess and understand the phylogenetic placement of NHP HBV lineages.

Origins and spread of genotypes A and D

The oldest ancient strains belonging to genotype A in our dataset are found in individuals from western Russia and the northern Caucasus dated between ~5 and ~4 ka (SGR004, RISE386/387, KBD002, KDC001; fig. S1). Around 2.5 ka, genotype A is observed in two Scythian individuals from the Pontic steppe (I0960) and the Carpathian basin (DA195), a region where it is still found later on in individuals dated to ~2.1 and ~1.5 ka, respectively (CHT001, DA119). As stated previously (14), these findings fit well with the known genetic relationship between western Scythians and Bronze Age populations of the western steppe (194), and suggest that the presence of this genotype in Africa today is due to later dispersal from Eurasia. The latter point is further supported by the phylogenetic position of a strain discovered in an Egyptian mummy dated around 1.9 ka (15) (Abusir1543), which appears basal to subgenotypes typically found in Africa today (A1 and A3; fig. S2). However, our results also reveal the presence of this genotype in northeastern Europe (BOO006/008) as well as in the Near East (ALA098/110, KEL045) between ~4 and ~3.5 ka, bringing additional mystery regarding its precise geographic origins and early dissemination routes.

Ancient HBV strains belonging to genotype D appear during the Iron Age in our dataset, with first occurrences dated around 2.5 ka in two individuals from the same site in today's Italian Alps (I0216 and I0217; fig. S1), as well as from different parts of the western steppe ranging from southwestern Russia to Central Asia (ORE002, AKB003, BRE026/028, DA51). From the early Medieval period on, this genotype is observed in large parts of Europe where it appears to have become predominant (Fig. 3), which it still is today (4). Although the geographical distribution of genotype D prior to the Iron Age cannot be directly assessed using our dataset, the apparent absence of this genotype in western Eurasia before that period, together with its wide presence in Central Asia, suggest an Asian origin followed by east-to-west expansions (Figs. 3, S1). However, recombination analyses indicate that this genotype might descend from a genotype A/WENBA hybrid, although this result should be viewed with caution (see "Recombination analysis" section; figs. S10, S5d).

Ancient genotype D strains from Europe do not form a monophyletic group in our phylogenetic reconstruction, which could suggest several waves of dissemination (fig. S2). Iron age genomes from Central Asia appear basal to most European Medieval genomes, which may reflect westward migrations across Eurasia occurring during the Migration Period (195). In contrast, earliest genotype D strains from the Italian Alps form the deepest genotype D branch, pointing to a distinct origin. Finally, several genotype D strains from Medieval Europe are found basal to subgenotypes which also prevail in other continents today (D3, and D4), suggesting that European colonization and modern migrations might have driven the spread of these lineages.



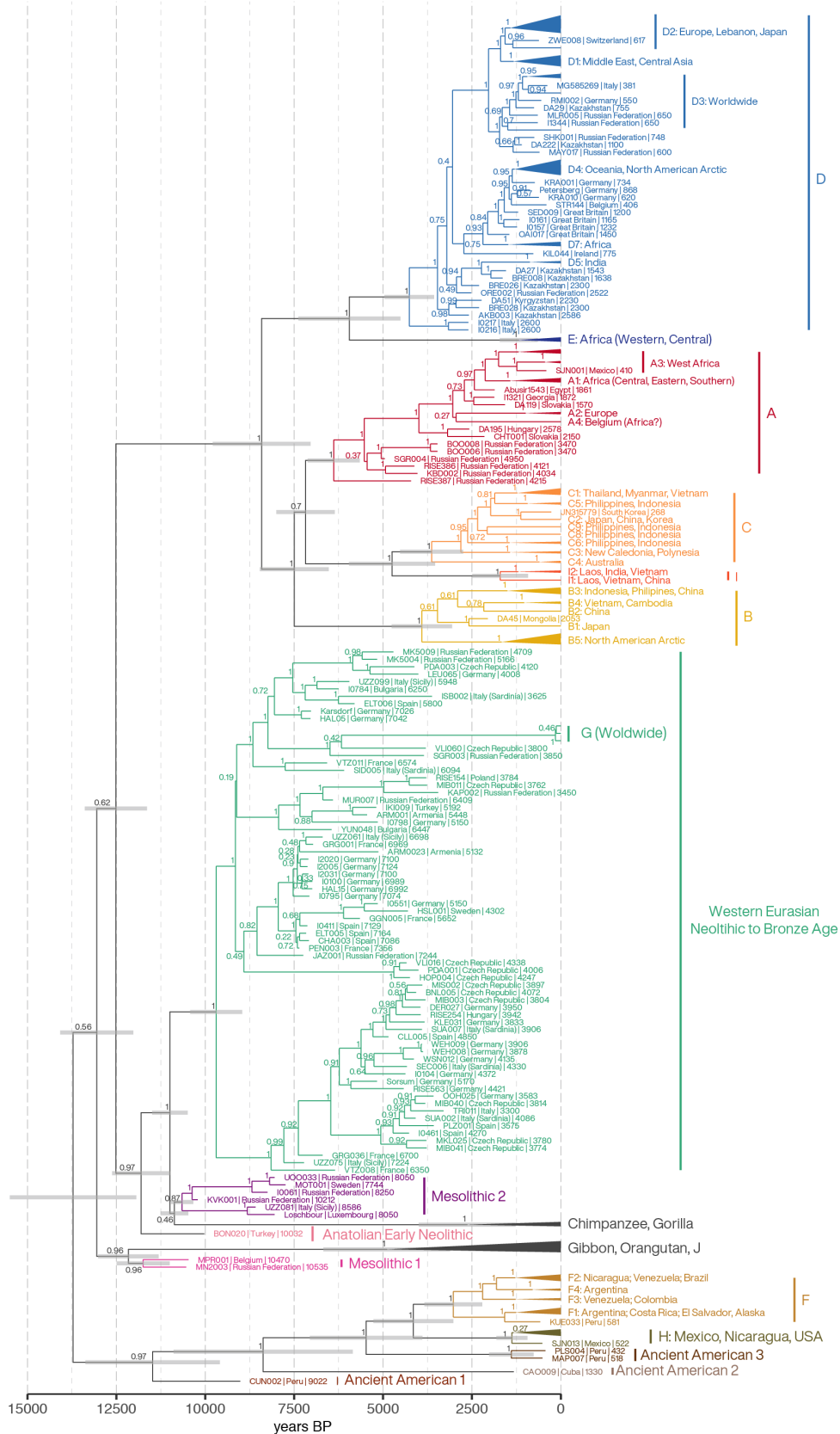


Figure S2. Time-calibrated phylogenetic tree of HBV. Sample labels are displayed for ancient genomes, together with their geographic locations and midrange date estimates. The main geographic distribution of each (sub)genotype is indicated. Posterior node support and age estimates (grey bars indicating 95% highest posterior density; except within genotypes) are reported.

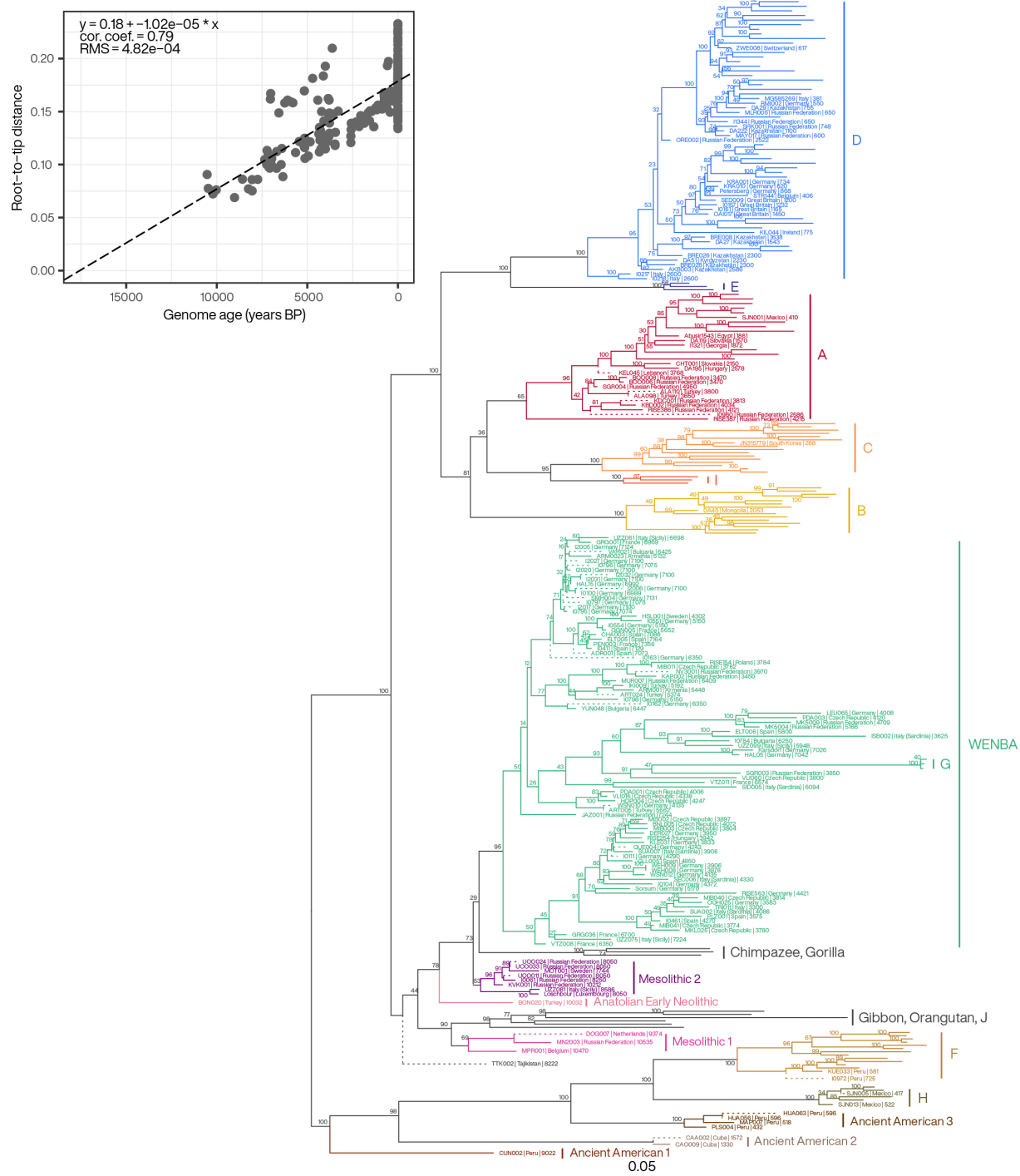


Figure S3. Phylogenetic tree with branches in substitutions per site estimated by maximum likelihood. The tree was midpoint-rooted. Bootstrap supports are reported on the nodes. The most likely phylogenetic position of low-covered genomes, as estimated with EPA-ng, are represented by dotted lines (note that these were not used for the construction of the tree). In the upper left panel: regression of root-to-tip genetic distances against tip dates.

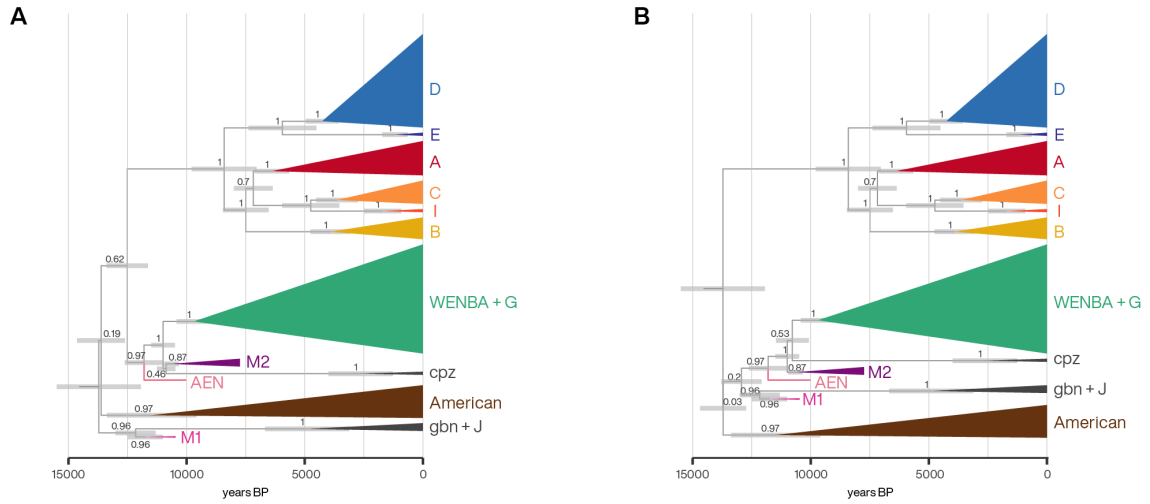


Figure S4. Time-calibrated phylogenetic trees of HBV representing alternative topologies for two low-supported clades: (A) the Eurasian branch and (B) the clade corresponding to the Eurasian branch with the exclusion of the Mesolithic 1 and Southeast Asian NHP clades. MCC trees conditioned on the absence of these clades were retrieved, and the full posterior tree sample was used to compute node posterior supports.

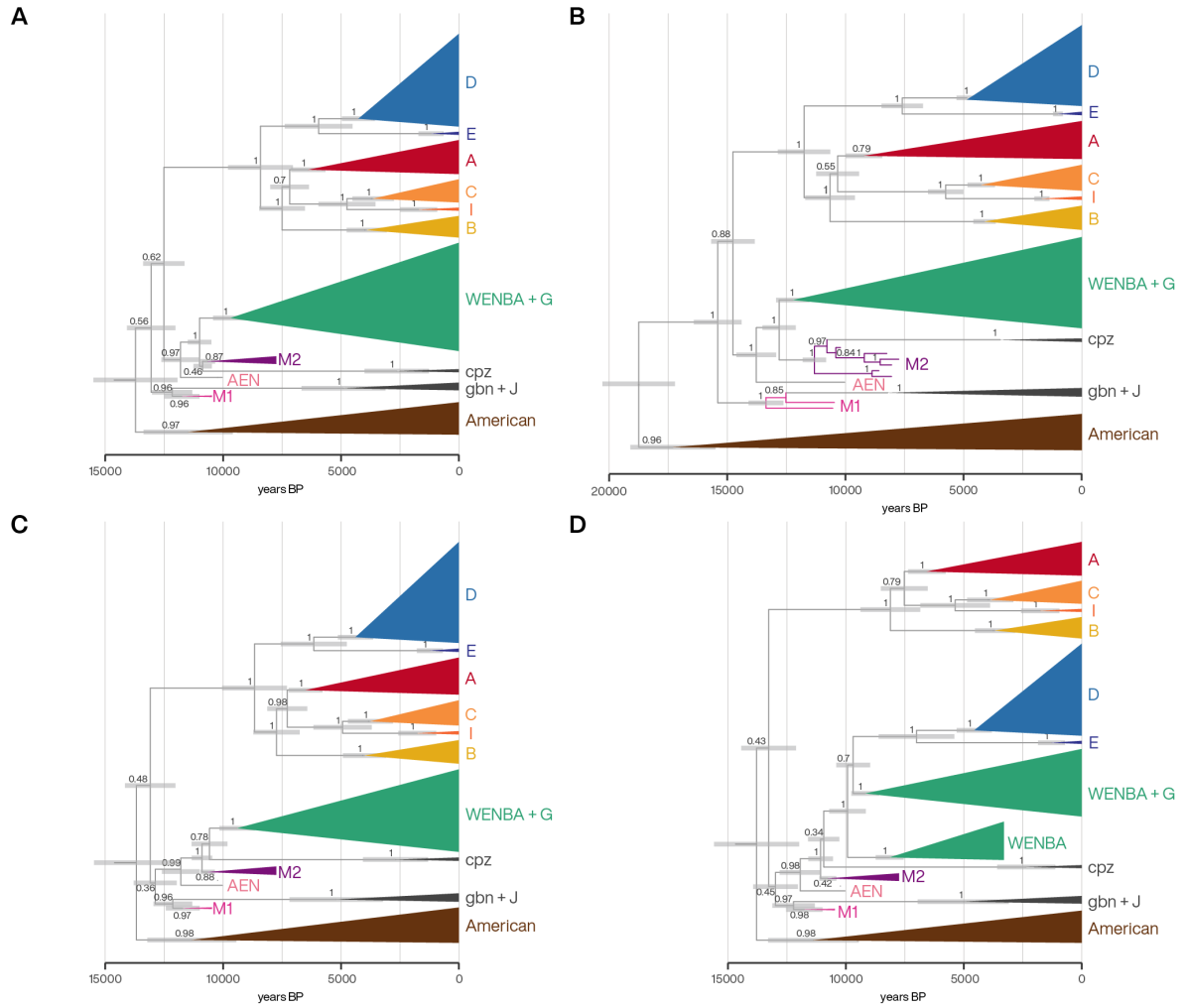


Figure S5. Time-calibrated phylogenetic trees of HBV reconstructed using different subsets of the data: (A) full dataset, (B) exclusion of modern genomes, (C) exclusion of genomes recovered from mixed infection contexts, (D) masking of recombining regions identified with RDP4. All analyses were performed using a skyline coalescent tree prior with five time intervals, and a lognormal relaxed clock.

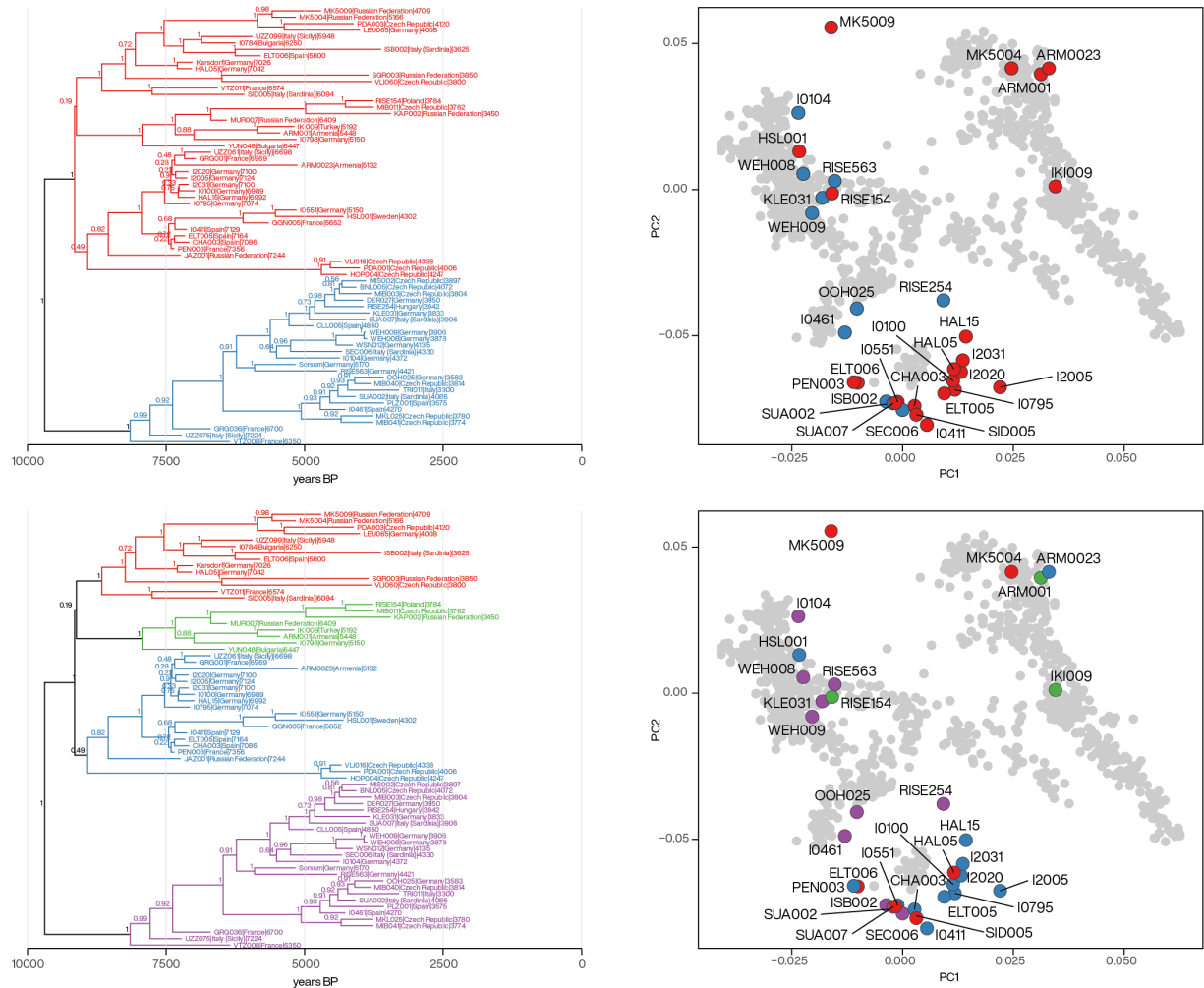


Figure S6. Exploration of potential correlations between the phylogenetic relationships of ancient HBV genomes within the WENBA clade and the genetic profile of the individuals carrying them. On the left: a subset of the time-calibrated phylogenetic tree corresponding to the WENBA clade (genotype G excluded) is colored according to two sublineage classification schemes. On the right, the principal components analysis plot of modern and ancient western Eurasians summarizes the genetic variation of a subset of individuals for which human genetic data was available. Individuals are colored according to the lineage of the HBV strain they carried, as in the tree on the left.

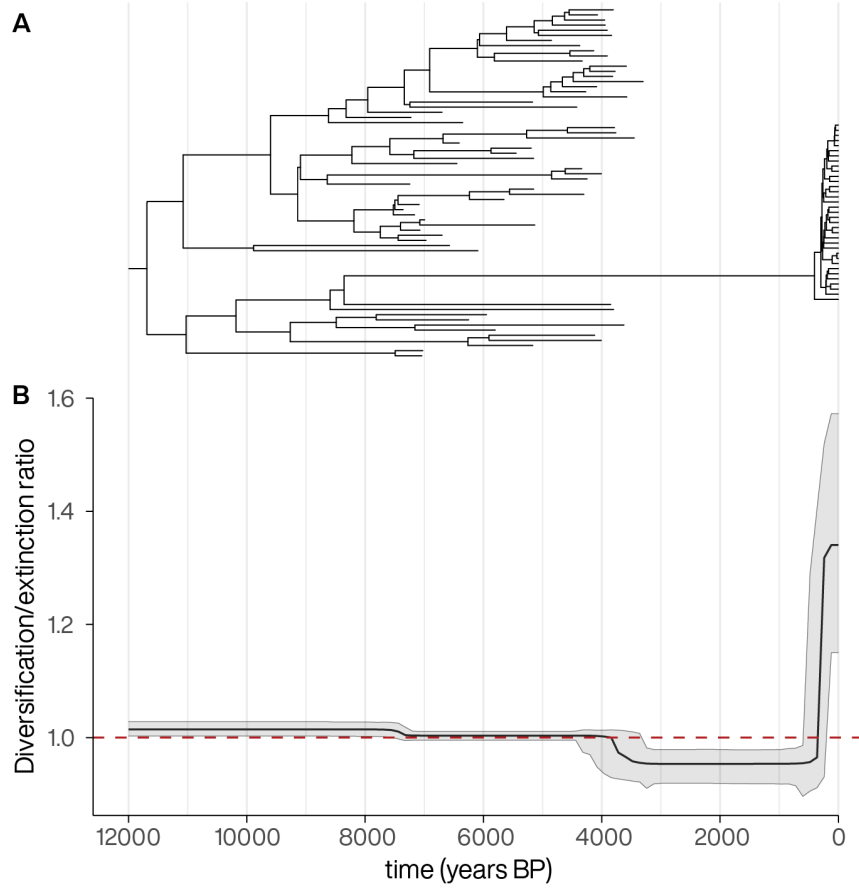


Figure S7. Phylodynamic analysis of the WENBA lineage. (A) Time-calibrated phylogenetic tree obtained using a birth death skyline tree prior and a strict clock model, with a dataset consisting of ancient WENBA strains and modern genotype G strains. (B) Skyline plot showing the estimated diversification/extinction ratio d through time with 95% HPD. The red dotted line indicates $d=1$, below and above which the number of infections decreases or increases, respectively.

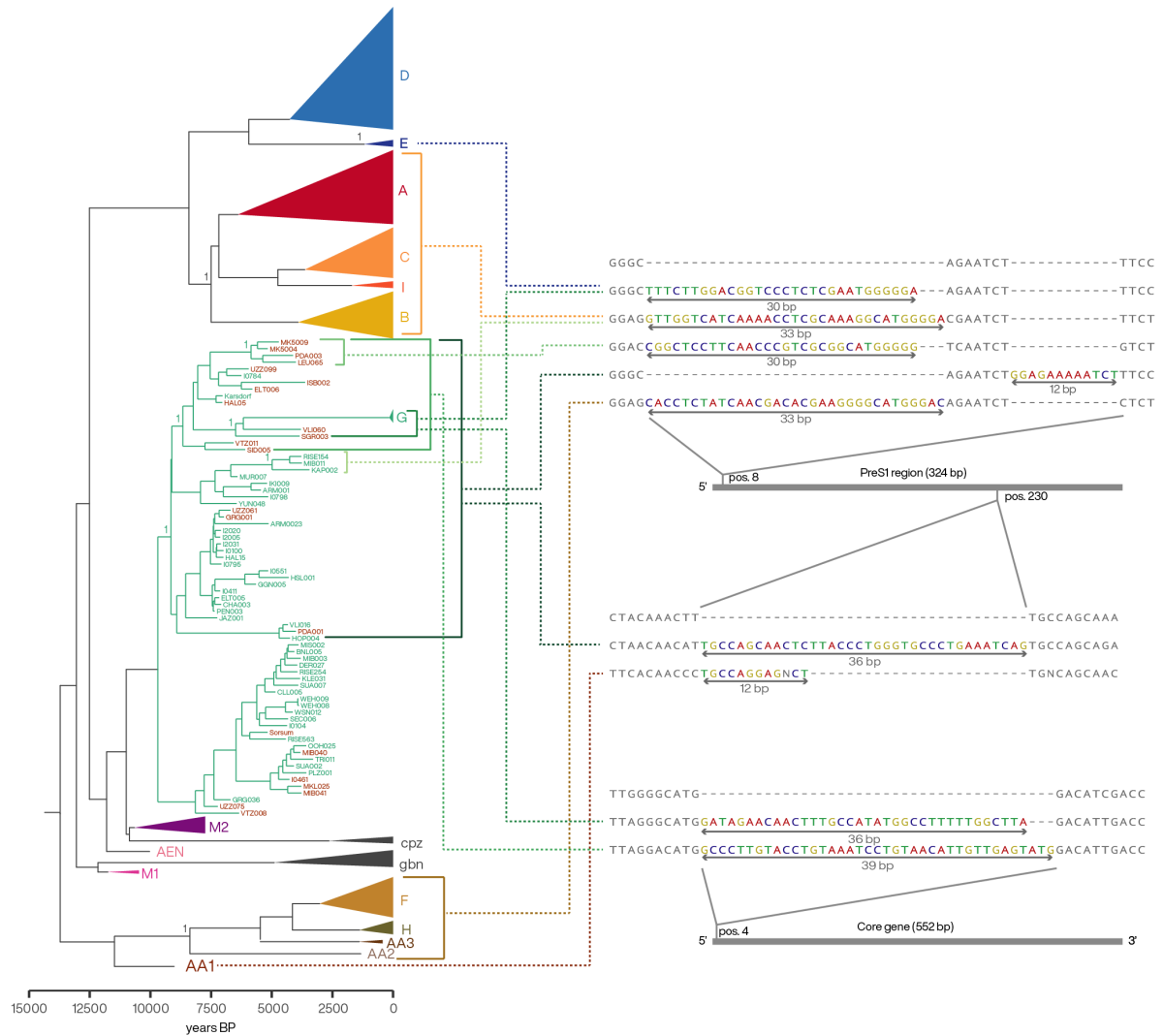


Figure S8. Insertions found in the core gene and PreS1 region of HBV genomes. For each genomic position in which insertions were found, an alignment of the main variants is shown. Insertions were aligned to facilitate visualization but might not represent homologous regions (they were masked for phylogenetic analyses). The first sequence corresponds to the short variant found in all genomes unless specified otherwise. Each insertion was found in entire monophyletic groups which are indicated by brackets on the time-calibrated phylogeny, together with their posterior support values. Genomes in the WENBA clade that were found in the context of mixed HBV infections are marked in brown.

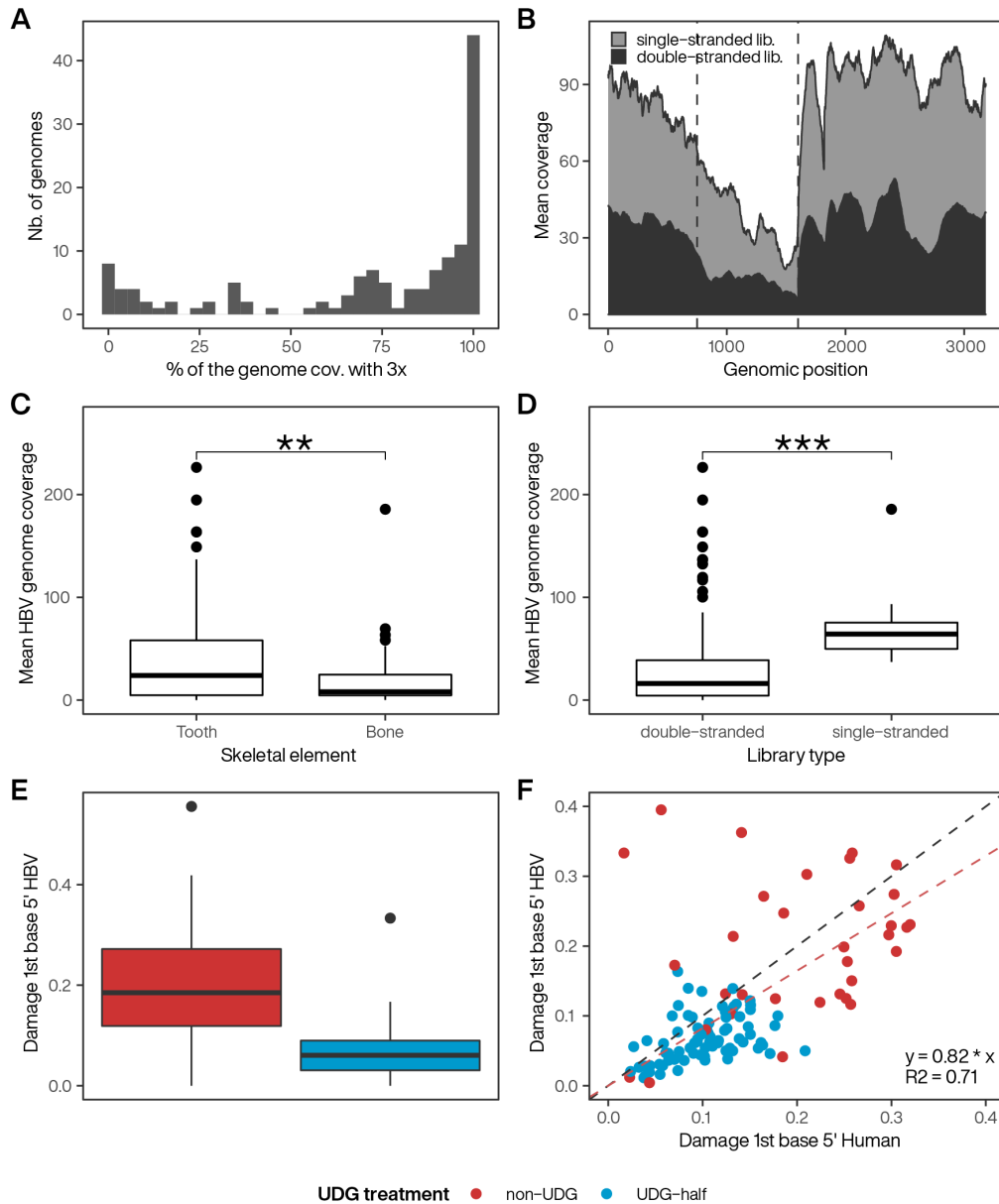


Figure S9. HBV genome coverage and damage statistics. (A) Histogram of the percentage of HBV genome recovered with $> 3x$ coverage. (B) Sequencing coverage across the HBV genome averaged over all libraries. Dotted lines represent approximate boundaries of the single-stranded region of the genome. (C,D) Boxplots of the mean coverage of HBV genomes recovered from each library, depending on the skeletal element from which DNA was extracted and the library protocol employed. Significance levels of pairwise mean comparisons based on a combined mixed linear model are represented with the following codes: * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$. (E) Boxplots of the frequency of damage-like substitutions on the 5' end of sequencing reads depending on the UDG treatment of libraries. (F) Scatter plot showing the correlation between the frequency of damage-like substitutions in HBV vs. Human-mapping reads across sequenced libraries. The identity line ($y=x$) is represented in black, and the fitted regression line in red.



Figure S10. RDP plots showing evidence of recombination events. The x-axes refer to the position in the sequence alignment. The y-axes refer to the pairwise identity of sequence pairs (restricted to phylogenetically-informative sites), as indicated by the color of each line. For each plot we indicate the identity of the sequence triplet that was used to infer the recombination event (R: recombinant, m: minor parent, M: major parent). Sequences that were used to infer a recombination event without being identified as one of the parental sequences are indicated by square brackets. When the same event was detected in several sequences, we report only the triplet that provided the strongest signal.

Table S1. Median estimates of HBV substitution rate and tMRCA obtained from the full dataset and different subsets of it.

Analysis	tMRCA (years BP; 95%HPD)	Substitution rate (sub.site⁻¹.years⁻¹; 95%HPD)
full dataset	13518 (12119-15680)	1.54e-05 (1.34e-05-1.74e-05)
mixed infections excluded	13497 (12080-15707)	1.45e-05 (1.26e-05-1.64e-05)
recombination-free alignment	13588 (12161-15844)	1.47e-05 (1.29e-05-1.67e-05)
time-dependent rate	18710 (17170-20240)	from 1.79e-05 (1.66e-05-1.93e-05) to 8.7e-06 (7.9e-06-9.7e-06)

Table S2. Results of low-coverage genome lineage assignments using EPA-ng.

ID	Country	Date	Taxonomic path	Assignment	LWR*
DOG007	Netherlands	9,374	HBV;Mesolithic 1	Mesolithic 1	0.81
TTK002	Tajikistan	8,222	HBV	Undetermined	0.93
UOO011	Russian	8,050	HBV;Mesolithic 2	Mesolithic 2	1.00
UOO024	Russian	8,050	HBV;Mesolithic 2	Mesolithic 2	0.98
SMH004	Germany	7,131	HBV;WENBA	WENBA	0.73
I2027	Germany	7,100	HBV;WENBA	WENBA	0.55
I2017	Germany	7,100	HBV;WENBA	WENBA	0.96
I2006	Germany	7,100	HBV;WENBA	WENBA	0.93
I2032	Germany	7,100	HBV;WENBA	WENBA	0.97
I0797	Germany	7,075	HBV;WENBA	WENBA	0.95
I0796	Germany	7,075	HBV;WENBA	WENBA	0.83
ADR001	Spain	7,073	HBV;WENBA	WENBA	0.99
VAR021	Bulgaria	6,425	HBV;WENBA	WENBA	0.84
I0163	Germany	6,350	HBV;WENBA	WENBA	0.87
I0162	Germany	6,350	HBV;WENBA	WENBA	0.93
ART005	Turkey	5,662	HBV;WENBA	WENBA	1.00
ART024	Turkey	5,374	HBV;WENBA	WENBA	0.98
I0554	Germany	5,150	HBV;WENBA	WENBA	1.00
I0111	Germany	4,290	HBV;WENBA	WENBA	0.98
QUE004	Germany	4,243	HBV;WENBA	WENBA	0.98
WSN010	Germany	4,135	HBV;WENBA	WENBA	0.99
NV3001	Russian	3,970	HBV;WENBA	WENBA	0.98
KDC001	Russian	3,813	HBV;A	A	0.99
ALA110	Turkey	3,800	HBV;A	A	0.92
KEL045	Lebanon	3,768	HBV;A	A	0.99
ALA098	Turkey	3,650	HBV;A	A	0.94
I0960	Russian	2,586	HBV;A	A	0.85
CAA002	Cuba	1,572	HBV;Ancient American 2	Ancient American	0.94
I0972	Peru	725	HBV;F	F	0.99
HUA056	Peru	596	HBV;Ancient American 3	Ancient American	1.00
HUA063	Peru	596	HBV;Ancient American 3	Ancient American	0.96
SJN005	Mexico	417	HBV;H	H	1.00

*LWR: likelihood weight ratio

Table S3. Results of lineage assignments of minor strains within mixed infections using EPA-ng.

ID	Country	Date (midrange estimate, years BP)	Lineage major strain	Assignment minor strain	LWR*
UZZ075	Italy (Sicily)	7,224	WENBA	WENBA	0.98
HAL05	Germany	7,042	WENBA	WENBA	0.73
GRG001	France	6,969	WENBA	WENBA	1.00
UZZ061	Italy (Sicily)	6,698	WENBA	WENBA	0.99
ELT006	Spain	5,800	WENBA	WENBA	0.94
Sorsum	Germany	5,170	WENBA	WENBA	0.13
I0461	Spain	4,270	WENBA	WENBA	0.78
LEU065	Germany	4,008	WENBA	WENBA	0.34
PDA001	Czech Republic	4,006	WENBA	WENBA	0.99
SGR003	Russian Federation	3,850	WENBA	WENBA	0.99
MKL025	Czech Republic	3,780	WENBA	WENBA	1.00
MIB041	Czech Republic	3,774	WENBA	WENBA	1.00

*LWR: likelihood weight ratio

Table S4. Minimum and maximum genetic distances between modern genotypes and different groups of ancient strains, together with the results of mPTP classification.

Genotype	mPTP class	A	Ancient A	B	C	D	Ancient D	E	G	I	gbn	cpz	M1	AEN	M2	WENBA
A	1	0/6	2.2/8.4	7.6/10.8	8.1/10.4	8.8/11.3	7.8/10.5	8.2/10.3	10/11	7/8.4	9.7/11.9	9.4/10.8	8/10.3	8/9.2	7.9/10.3	7/11.6
Ancient A	1,2	2.2/8.4	0/7.2	6.8/10	6.5/9.1	7.1/10.5	5.6/9.3	7.5/9.2	8.2/10.3	5.4/8.1	8.6/11.1	8/9.9	6.5/8.6	6.2/8.1	6.1/8.4	5.5/10.7
B	3	7.6/10.8	6.8/10	0/7.2	7.8/11.3	9.2/11.4	8.5/11	9.2/11.3	11.2/12.1	8/10.2	9.6/11.5	9.6/11.2	8/10.6	8/9.3	8/10.5	6.9/11.8
C	4	8.1/10.4	6.5/9.1	7.8/11.3	0/7.8	9/11.4	8/10.6	8.8/10.9	11.1/12.6	6.4/8	9.3/11	8.7/10.9	7.1/9.6	6.9/8.2	7.4/9.4	7/11.4
D	5	8.8/11.3	7.1/10.5	9.2/11.4	9/11.4	0/6.4	1.2/6.2	6.8/8.9	10.7/11.9	9/10.5	10.2/12.3	9.9/11.4	8.7/10.8	7.4/8.8	7.5/10	6.4/11.3
Ancient D	5	7.8/10.5	5.6/9.3	8.5/11	8/10.6	1.2/6.2	0/4.4	6.3/7.8	10/11	7.8/9.6	9.4/11.7	8.5/10.4	7.4/9.5	6.5/7.8	6.4/8.8	5.2/10.6
E	6	8.2/10.3	7.5/9.2	9.2/11.3	8.8/10.9	6.8/8.9	6.3/7.8	0/2.3	10.3/10.6	9/9.6	9.9/10.8	8.8/9.9	8.2/8.9	7.6/7.9	7.1/8.1	5.7/10
G	7	10/11	8.2/10.3	11.2/12.1	11.1/12.6	10.7/11.9	10/11	10.3/10.6	0/0.3	9.8/10.2	11/12.5	10.6/11.1	10/10.5	7.8/7.9	8.7/9.4	7/11
I	4	7/8.4	5.4/8.1	8/10.2	6.4/8	9/10.5	7.8/9.6	9/9.6	9.8/10.2	0/3.3	9.7/10.5	9.4/9.9	7.5/8.3	7.2/7.6	7.1/7.9	6.4/10.4
gbn	8	9.7/11.9	8.6/11.1	9.6/11.5	9.3/11	10.2/12.3	9.4/11.7	9.9/10.8	11/12.5	9.7/10.5	0/9.3	8.9/10.2	5.6/7.5	5.5/7.3	5.9/9.2	6.6/11.3
cpz	9	9.4/10.8	8/9.9	9.6/11.2	8.7/10.9	9.9/11.4	8.5/10.4	8.8/9.9	10.6/11.1	9.4/9.9	8.9/10.2	0/5.4	6.5/7.8	6.2/6.5	5/6.2	5.2/9.6
M1	8	8/10.3	6.5/8.6	8/10.6	7.1/9.6	8.7/10.8	7.4/9.5	8.2/8.9	10/10.5	7.5/8.3	5.6/7.5	6.5/7.8	0/3.1	3.2/4.6	3/5.2	3.9/9.8
AEN	10	8/9.2	6.2/8.1	8/9.3	6.9/8.2	7.4/8.8	6.5/7.8	7.6/7.9	7.8/7.9	7.2/7.6	5.5/7.3	6.2/6.5	3.2/4.6	0/0	2.7/3.4	2.8/6.7
M2	11	7.9/10.3	6.1/8.4	8/10.5	7.4/9.4	7.5/10	6.4/8.8	7.1/8.1	8.7/9.4	7.1/7.9	5.9/9.2	5/6.2	3/5.2	2.7/3.4	0/2.6	1.9/7.7
WENBA	12-20	7/11.6	5.5/10.7	6.9/11.8	7/11.4	6.4/11.3	5.2/10.6	5.7/10	7/11	6.4/10.4	6.6/11.3	5.2/9.6	3.9/9.8	2.8/6.7	1.9/7.7	0/8.2

*M1/2: Mesolithic 1/2; AEN: Anatolian Early Neolithic; WENBA: Western-Eurasian Neolithic-to-Bronze-Age

Table S5. Results of phylogenetic model selection using path sampling

tree prior	clock model	Bayes factor (log)*
Skyline coalescent with 5 time intervals	Relaxed lognormal	0
Birth-death skyline	Relaxed lognormal	-36
Skyline coalescent with 10 time intervals	Relaxed lognormal	-44
Exponential coalescent	Relaxed lognormal	-80
Constant coalescent	Relaxed lognormal	-153
Birth-death skyline	Strict	-431
Skyline coalescent with 10 time intervals	Strict	-433
Skyline coalescent with 5 time intervals	Strict	-435
Exponential coalescent	Strict	-491
Constant coalescent	Strict	-572

* The Bayes factors reported here are relative to the model yielding the highest marginal likelihood

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